US 20250287532A1

(19) United States
(12) Patent Application Publication
Lau

(10) Pub. No.: US 2025/0287532 A1
(43) Pub. Date: Sep. 11, 2025

(54) **MODULAR DATA CENTER**

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(21) Appl. No.: **19/216,525**

(22) Filed: **May 22, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/101,872, filed on Jan. 26, 2023, now Pat. No. 12,342,502, which is a continuation of application No. PCT/EP2021/055404, filed on Mar. 3, 2021, which is a continuation of application No. 16/939,570, filed on Jul. 27, 2020, now Pat. No. 10,966,349.

Publication Classification

(51) **Int. Cl.**

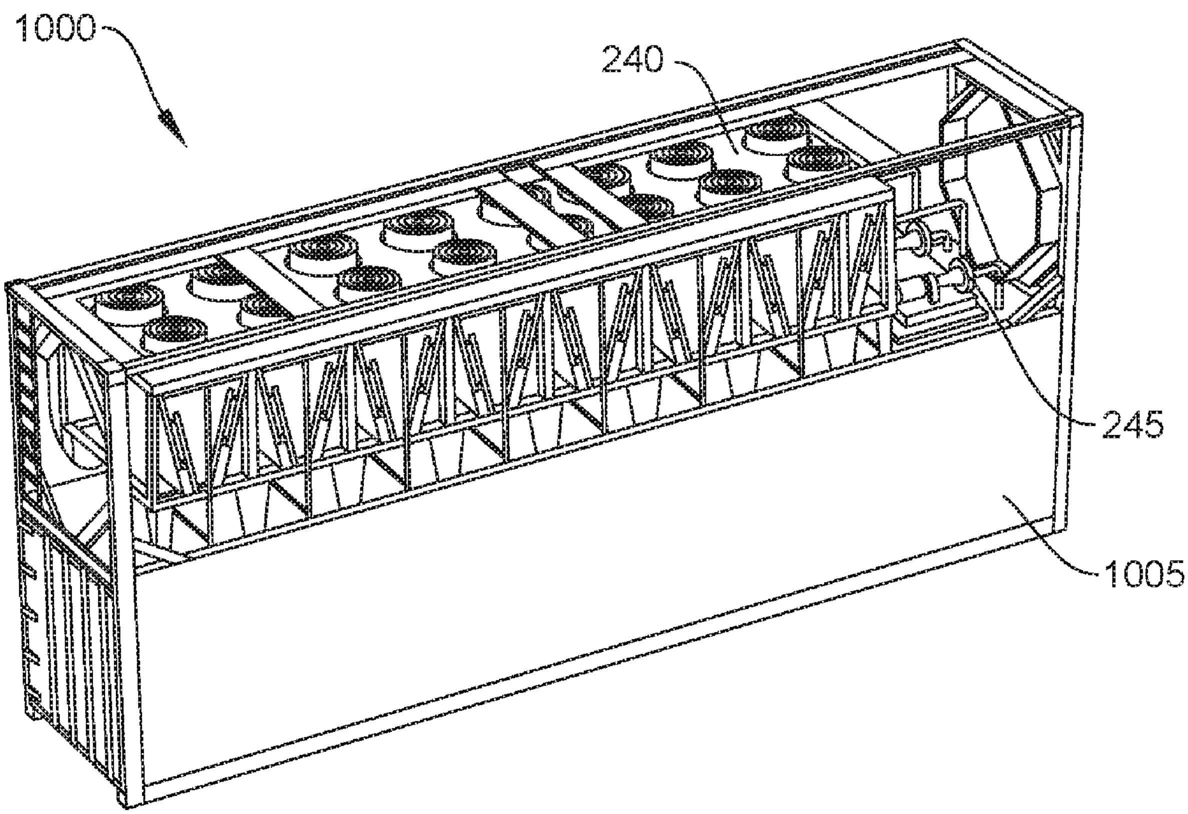
H05K 7/20	(2006.01)
F28F 27/02	(2006.01)
G01F 1/74	(2006.01)

(52) **U.S. Cl.**

CPC ***H05K 7/20281*** (2013.01); ***F28F 27/02*** (2013.01); ***G01F 1/74*** (2013.01); ***H05K 7/20236*** (2013.01); ***H05K 7/203*** (2013.01); ***H05K 7/20318*** (2013.01); ***H05K 7/20327*** (2013.01); ***H05K 7/20818*** (2013.01)

(57) **ABSTRACT**

A modular data center can include a shipping container housing an immersion tank having a primary condenser in thermal communication with an interior volume of the immersion tank and a vapor management system fluidically connected to the immersion tank. The vapor management system can enable the apparatus to effectively manage periods of high vapor production by removing vapor and other gases from a headspace of the immersion tank, condensing the vapor to liquid, and returning the liquid to the immersion tank. The modular data center can include an external heat rejection system mounted outside the shipping container and fluidly connected to the primary condenser.



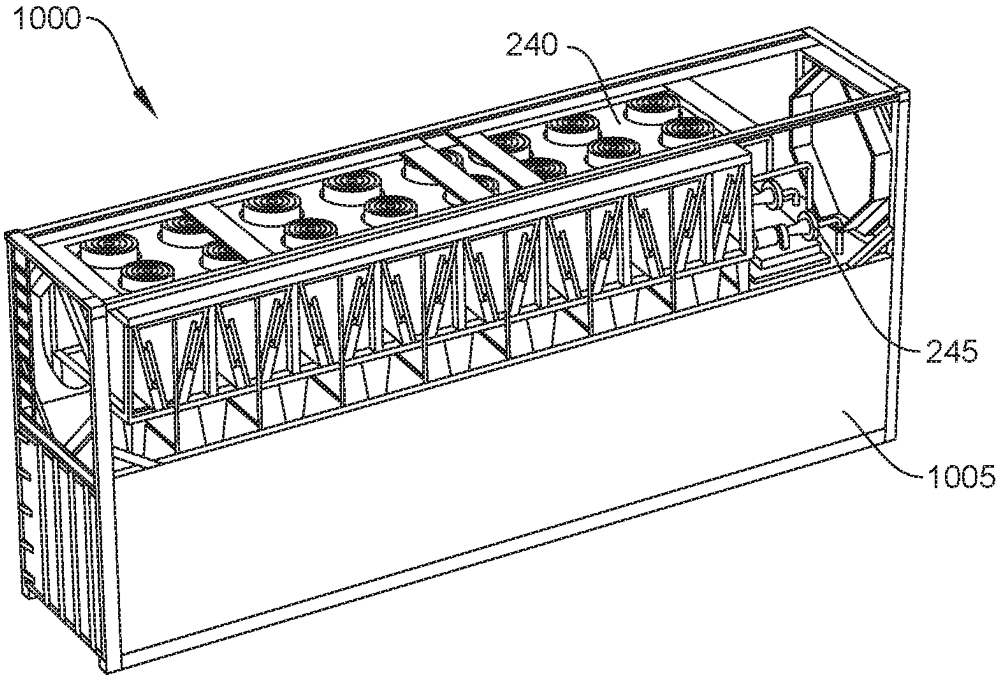


FIG. 1

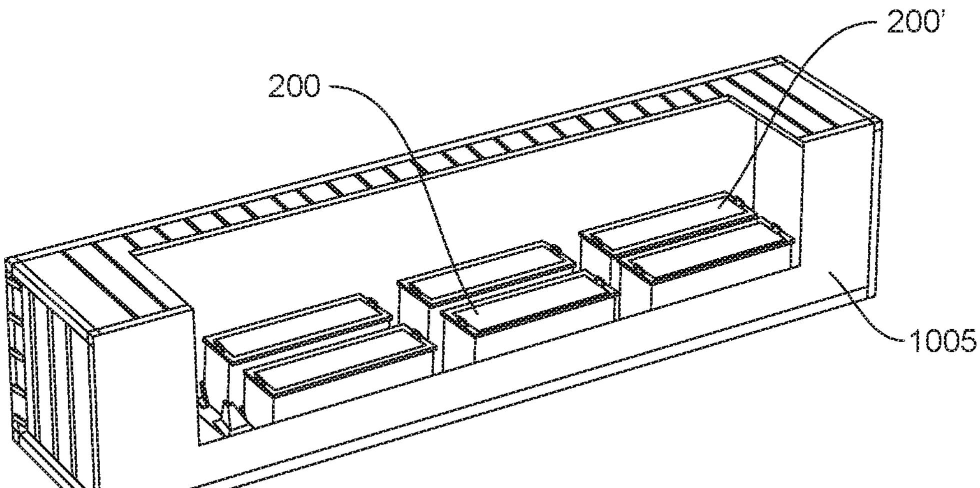


FIG. 2

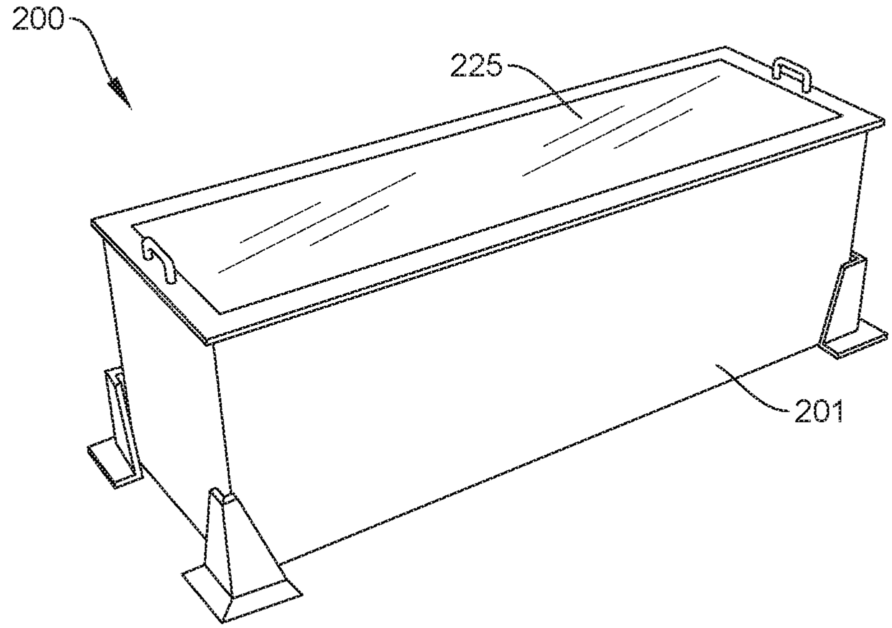
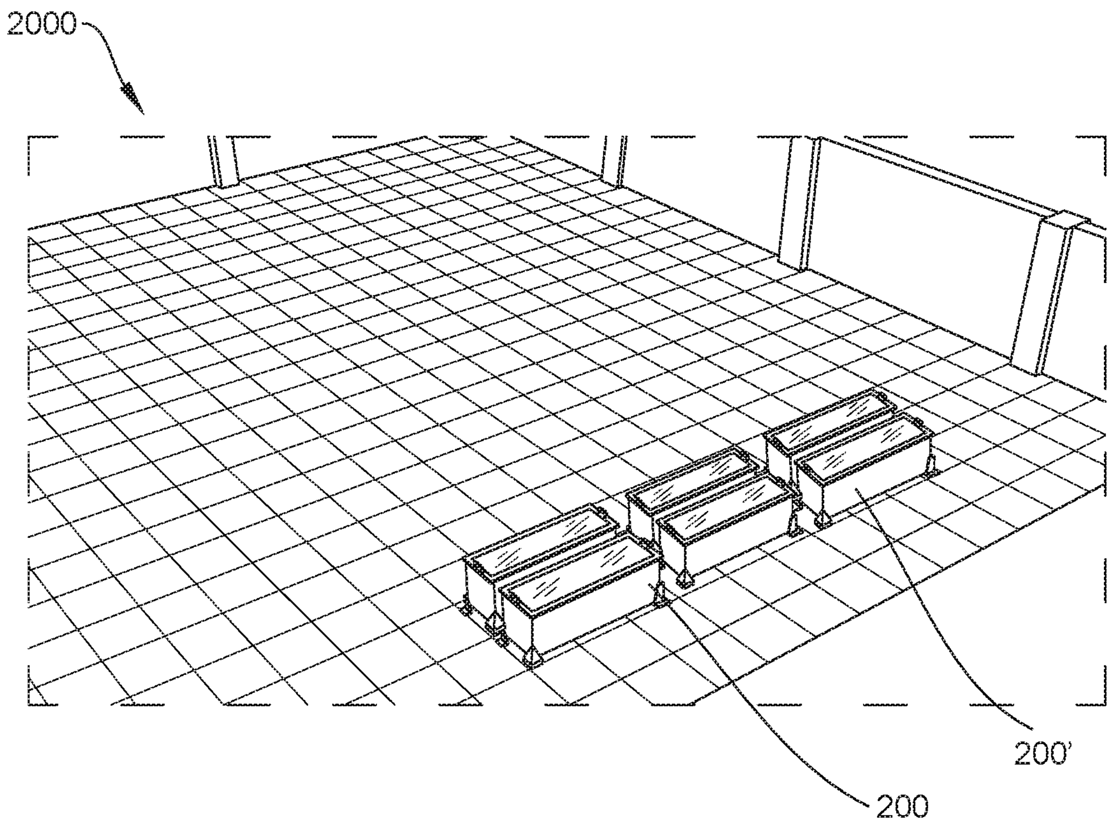


FIG. 3



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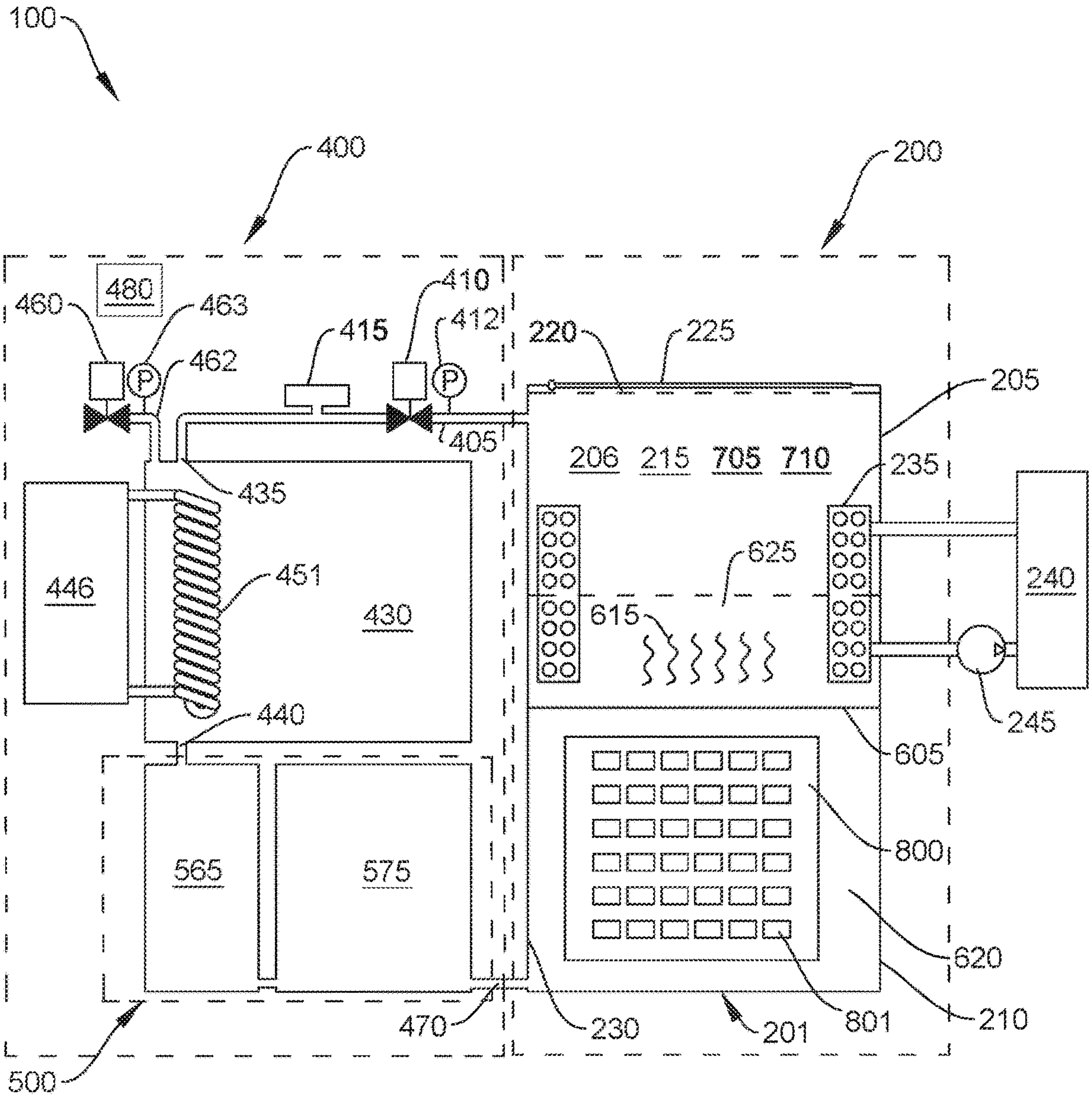
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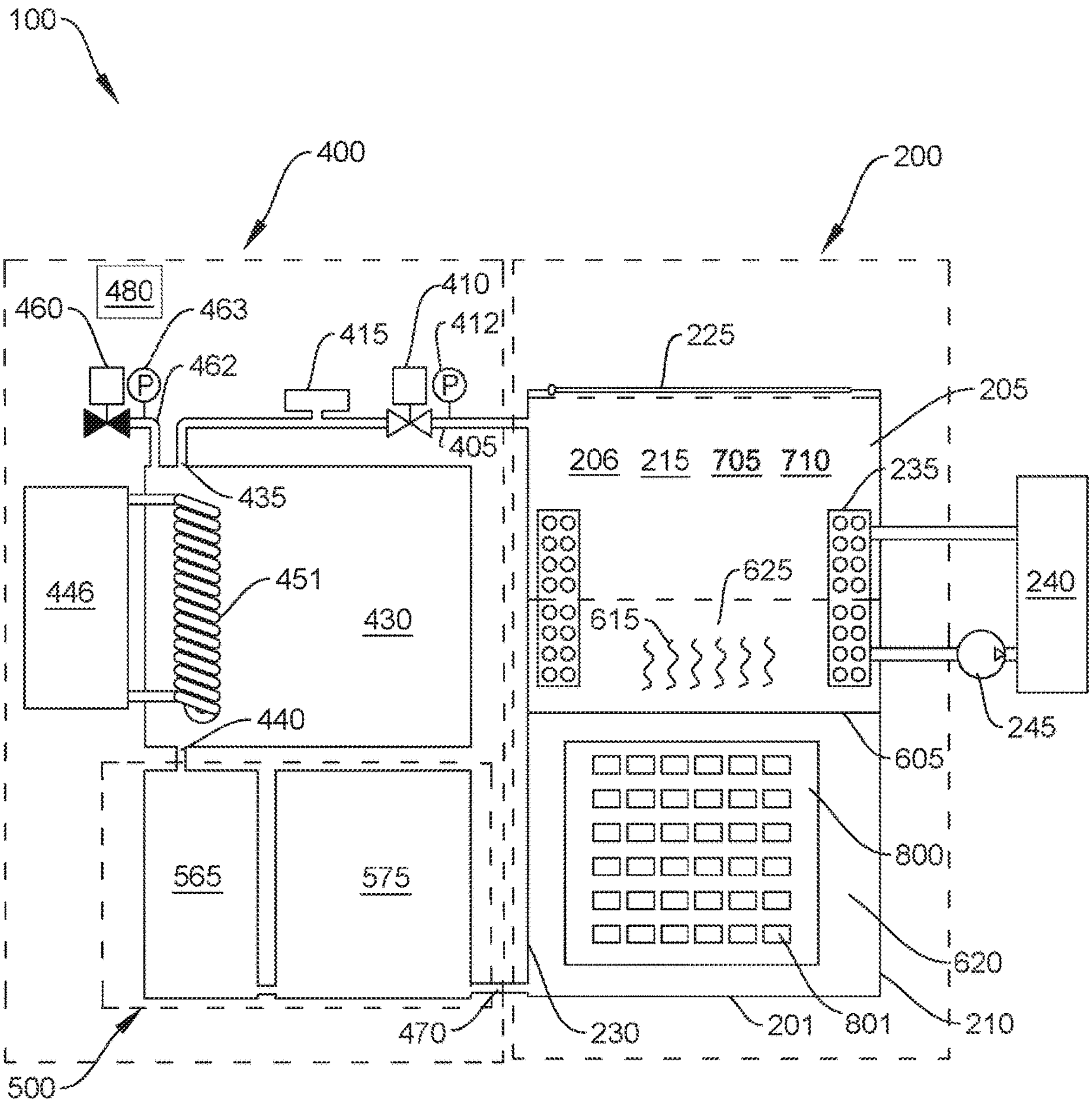
FIG. 5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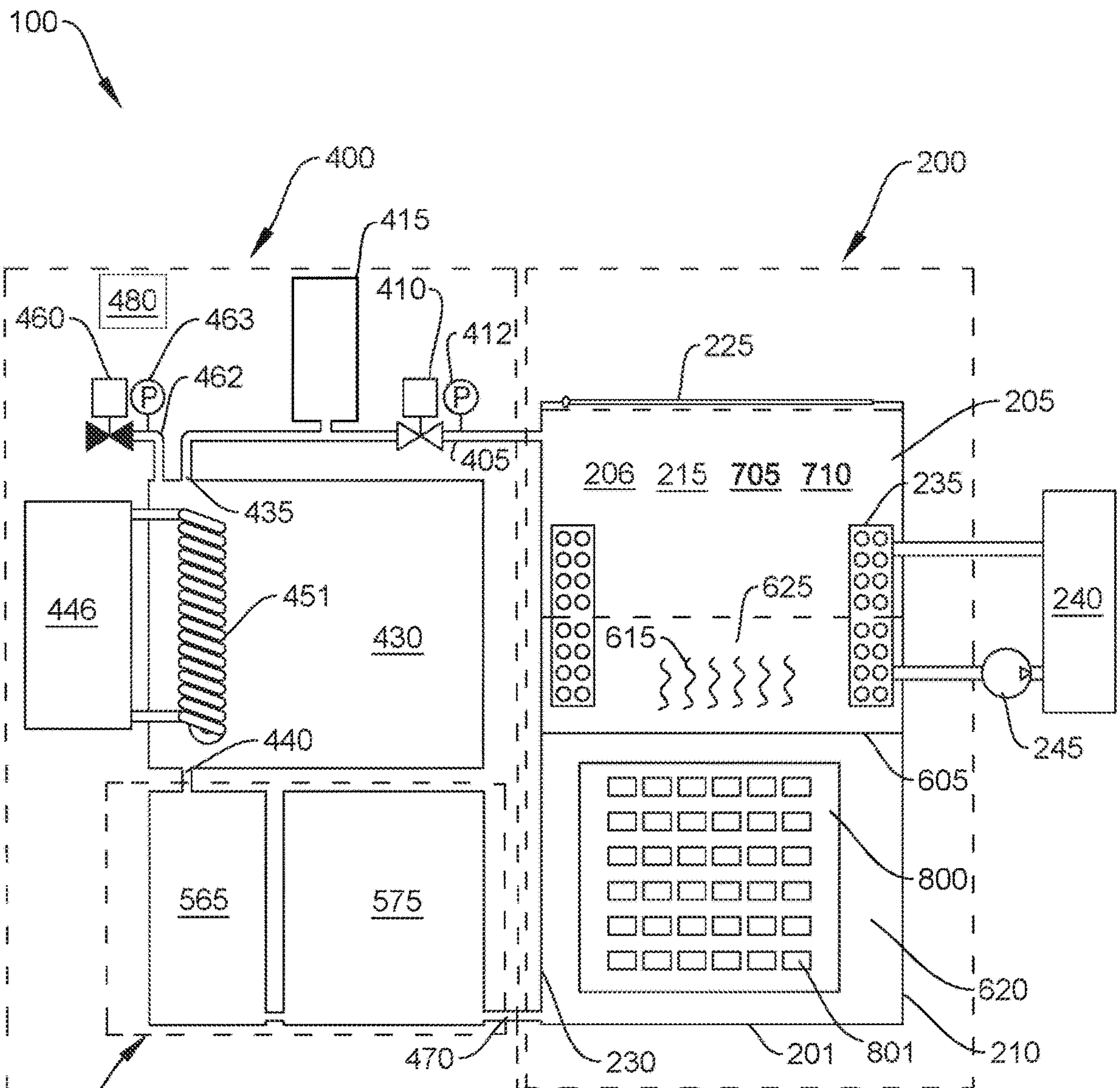
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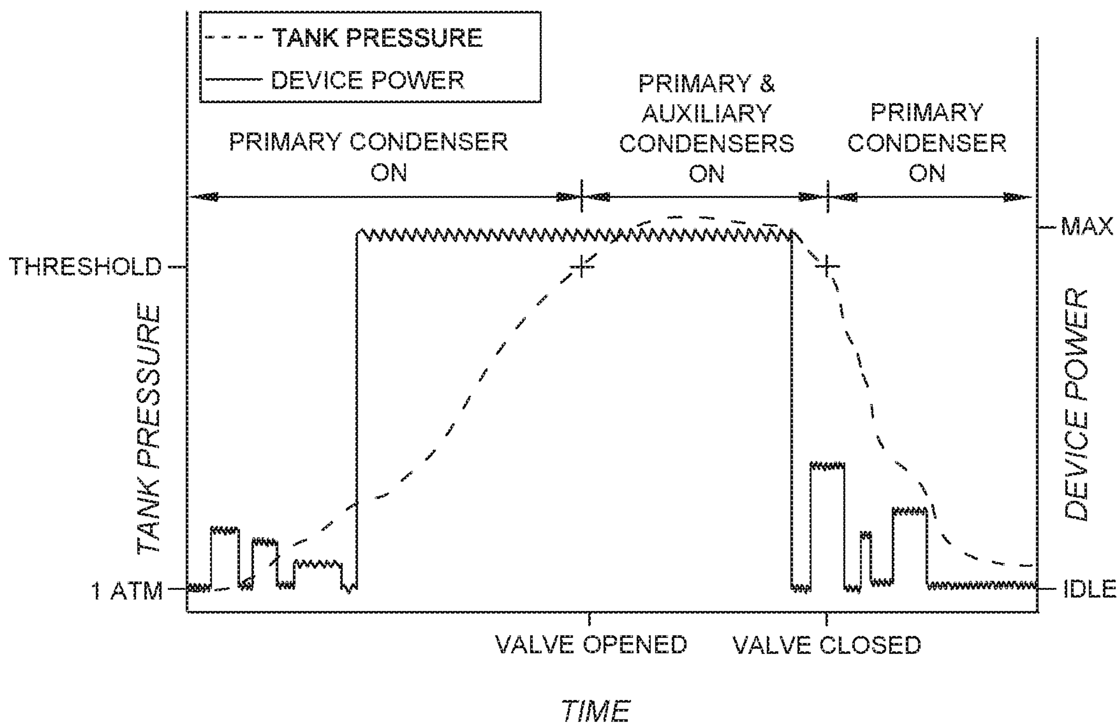
FIG. 7



TANK PRESSURE
DEVICE POWER
PRIMARY CONDENSER ON
PRIMARY & AUXILIARY CONDENSERS ON
PRIMARY CONDENSER ON
THRESHOLD
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MAX
IDLE
TANK PRESSURE
DEVICE POWER
VALVE OPENED
VALVE CLOSED
TIME

FIG. 8

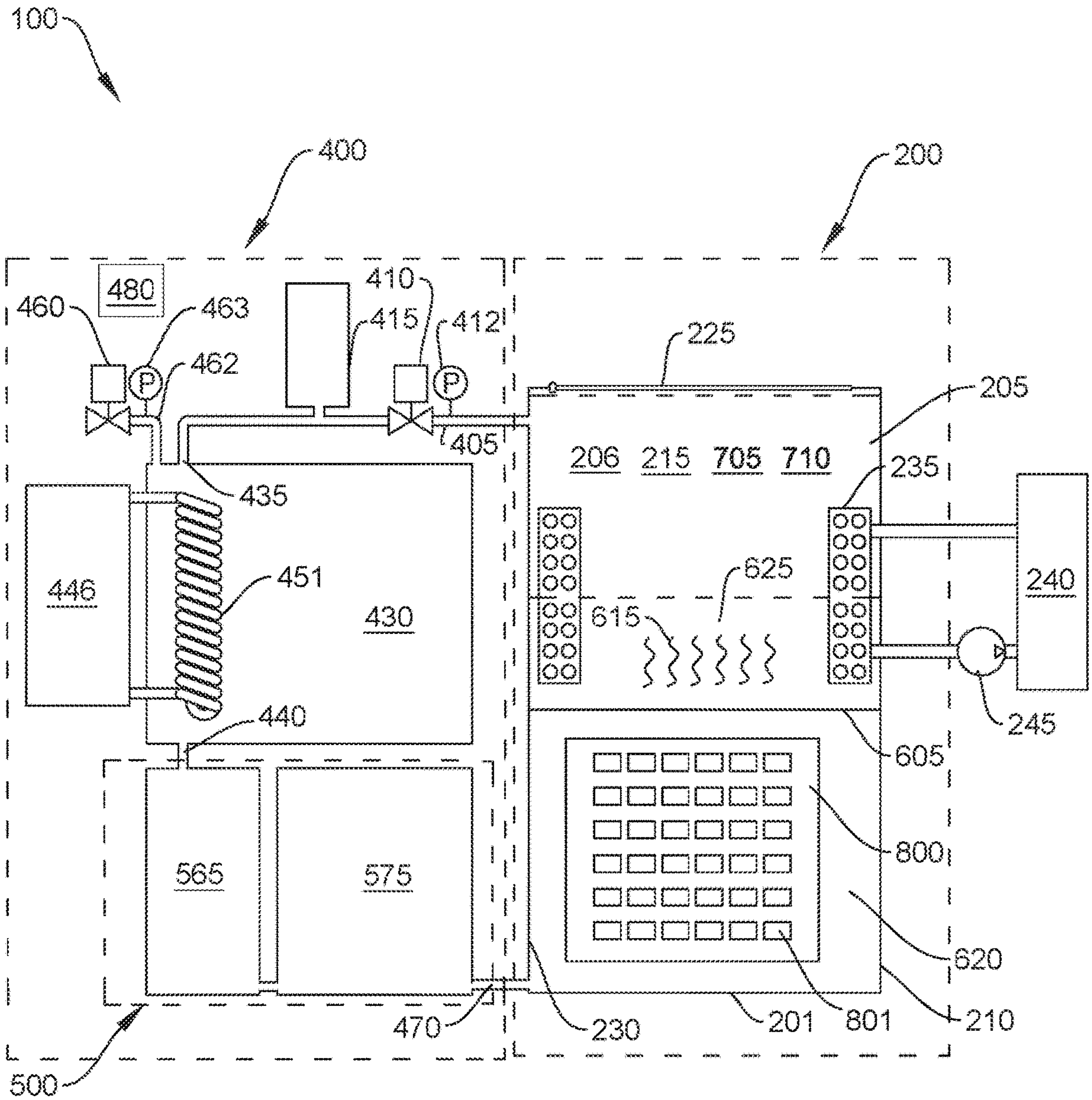
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FIG. 9

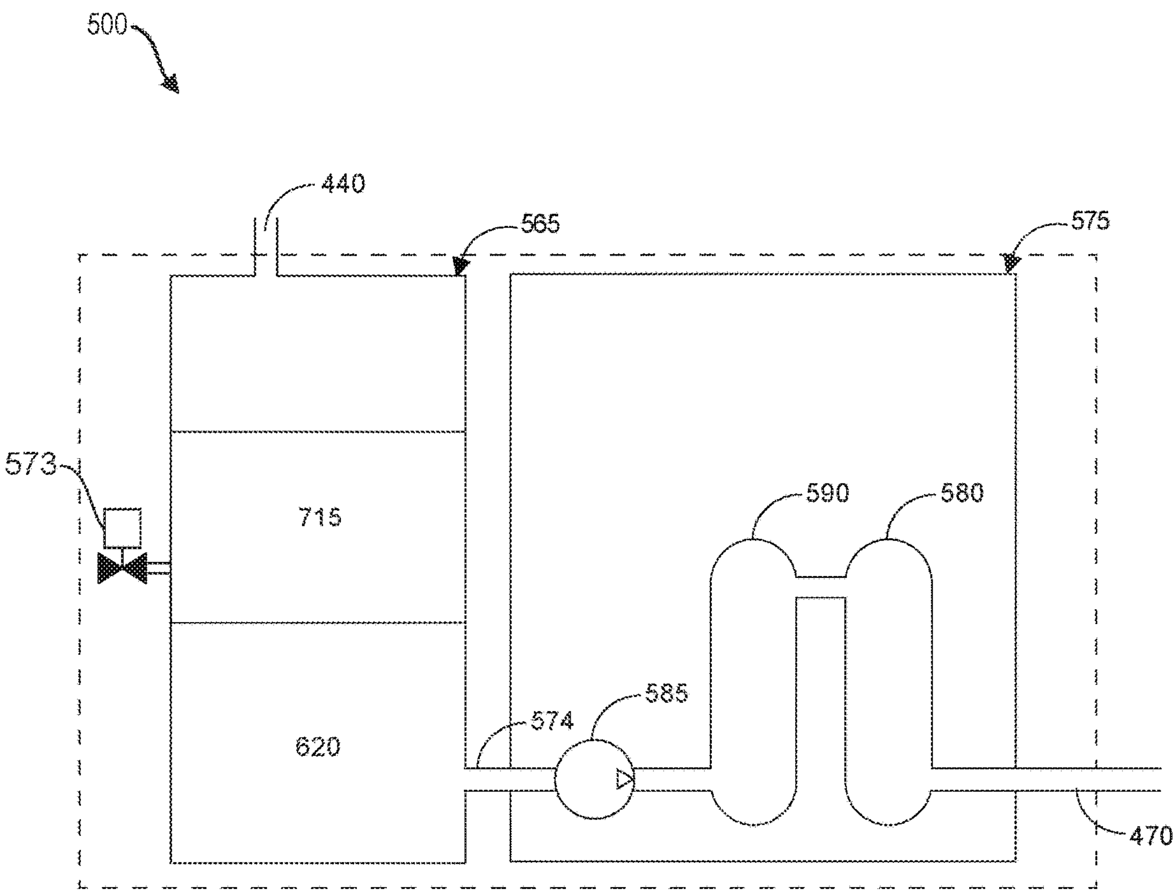
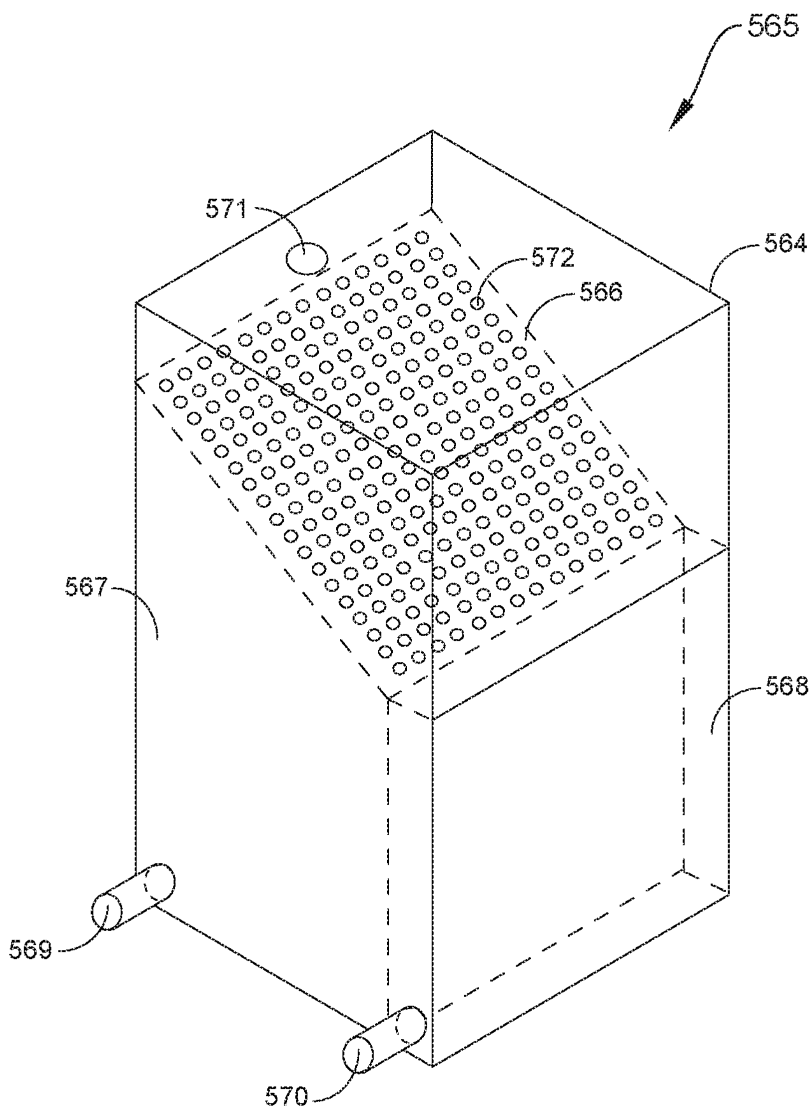


FIG. 10



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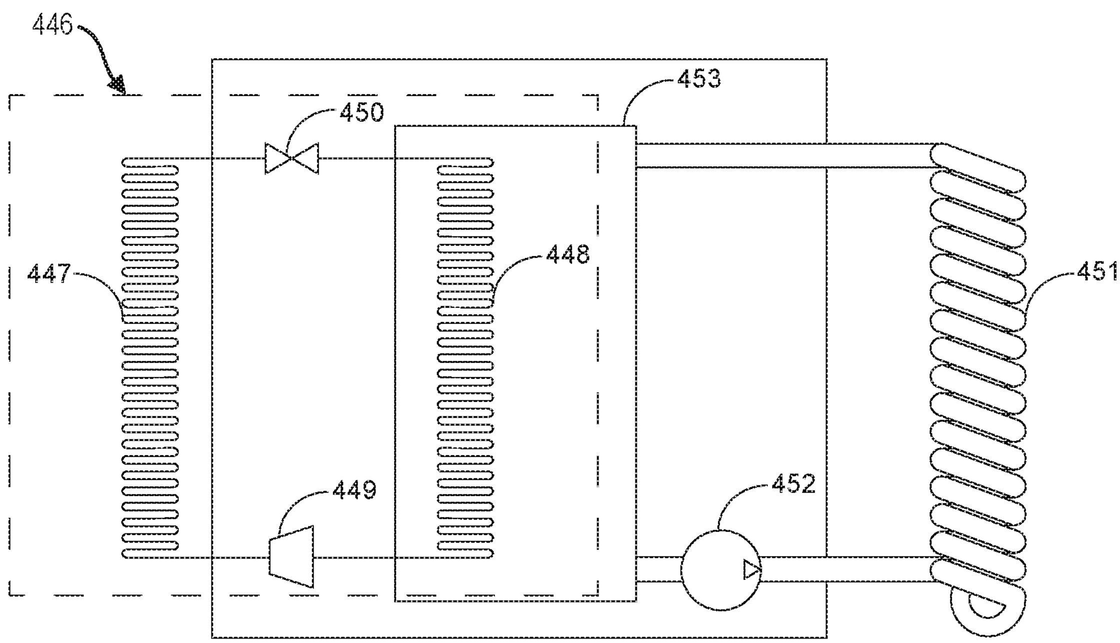

FIG. 12

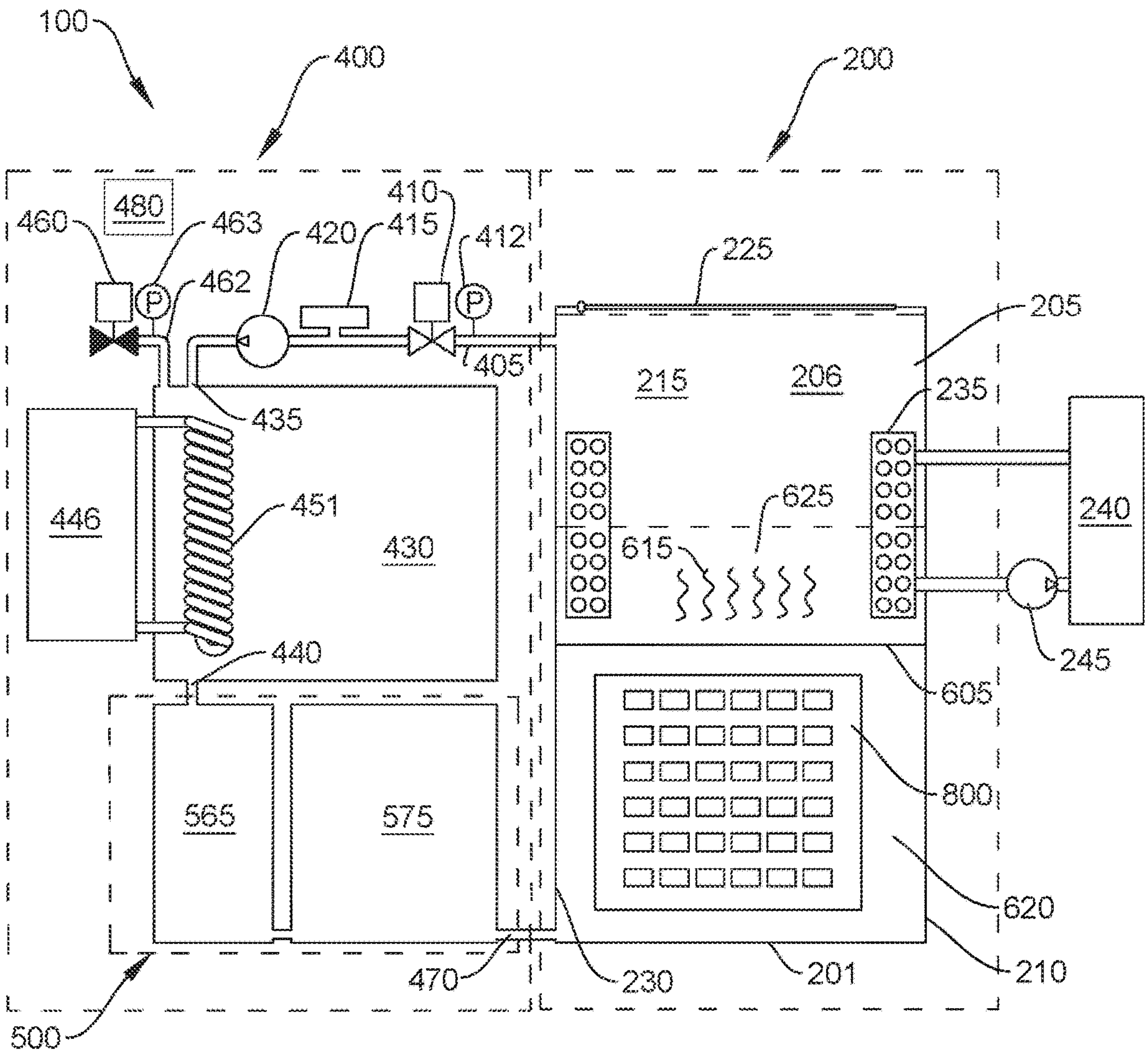
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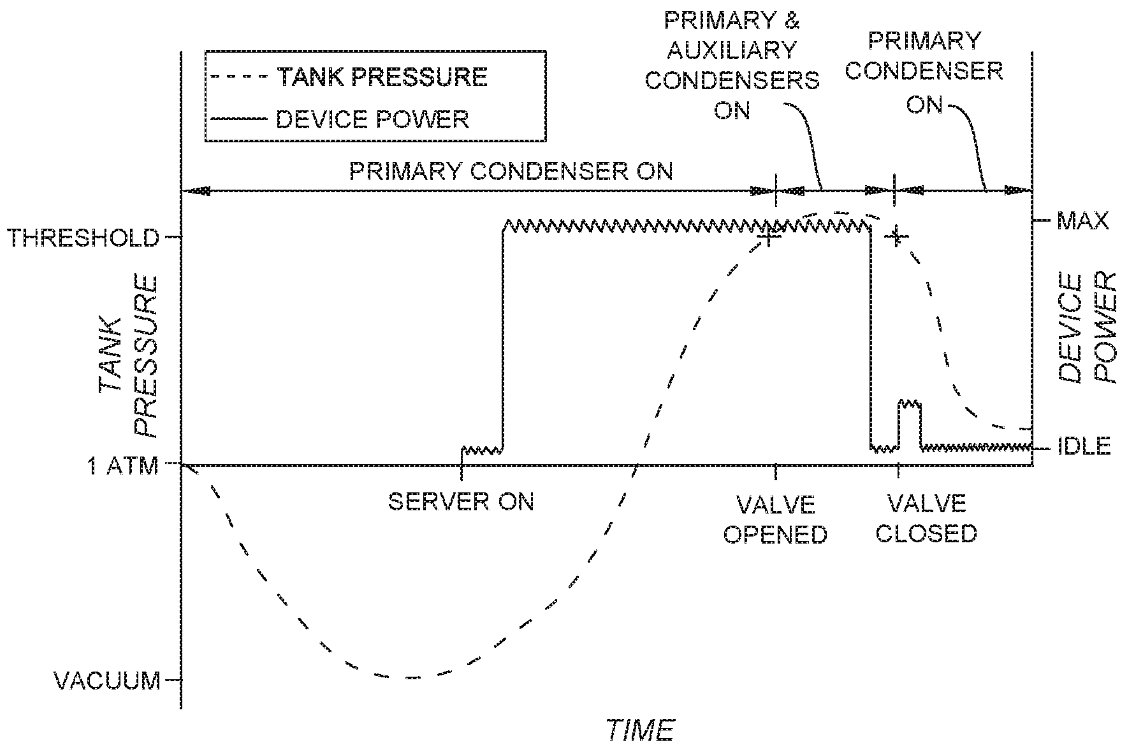
FIG. 13



TANK PRESSURE
DEVICE POWER
PRIMARY & AUXILIARY CONDENSERS ON
PRIMARY CONDENSER ON
PRIMARY CONDENSER ON
THRESHOLD
MAX
TANK PRESSURE
DEVICE POWER
1 ATM
IDLE
SERVER ON
VALVE OPENED
VALVE CLOSED
VACUUM
TIME

FIG. 14

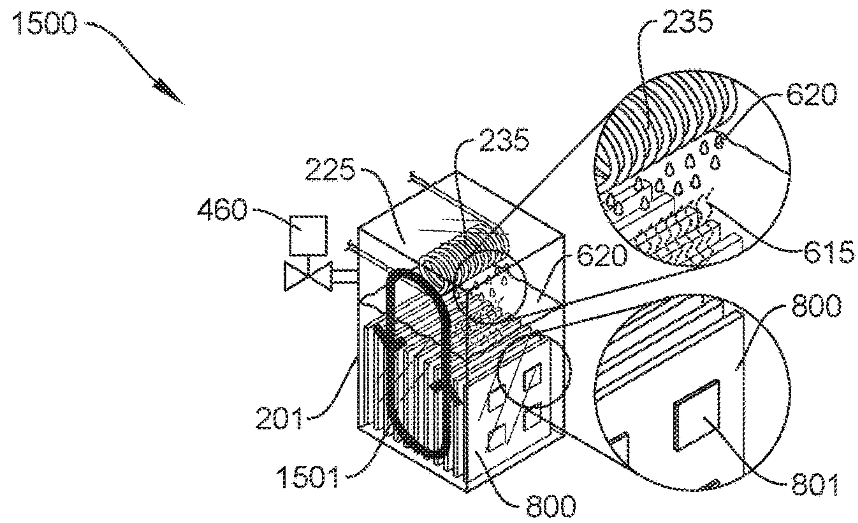


FIG. 15 (PRIOR ART)

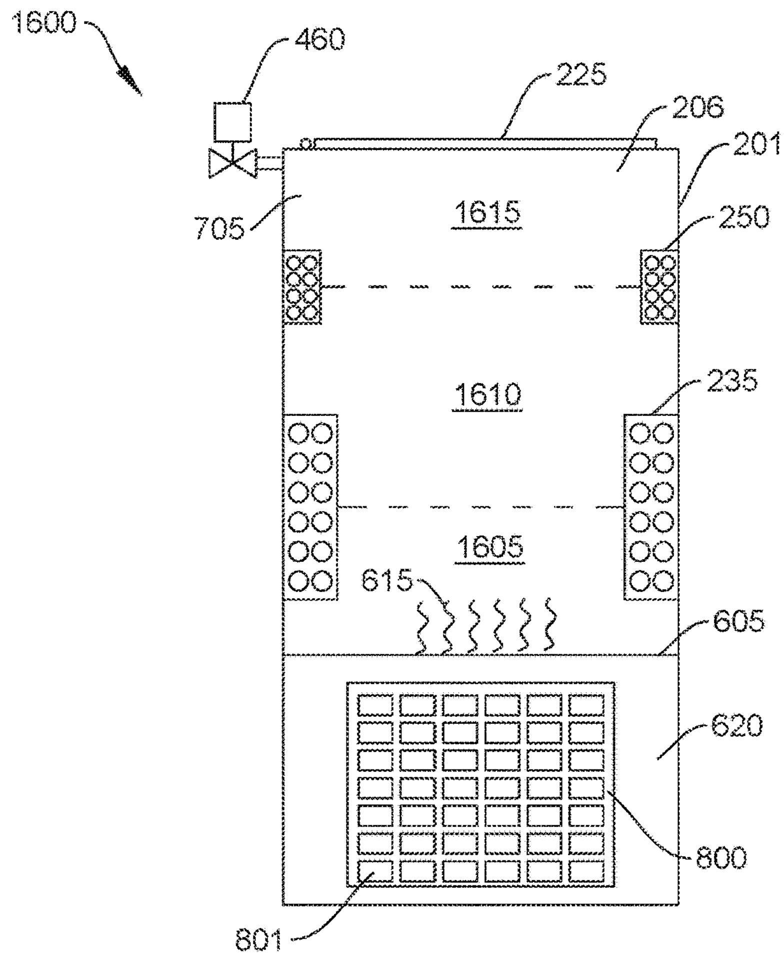


FIG. 16 (PRIOR ART)

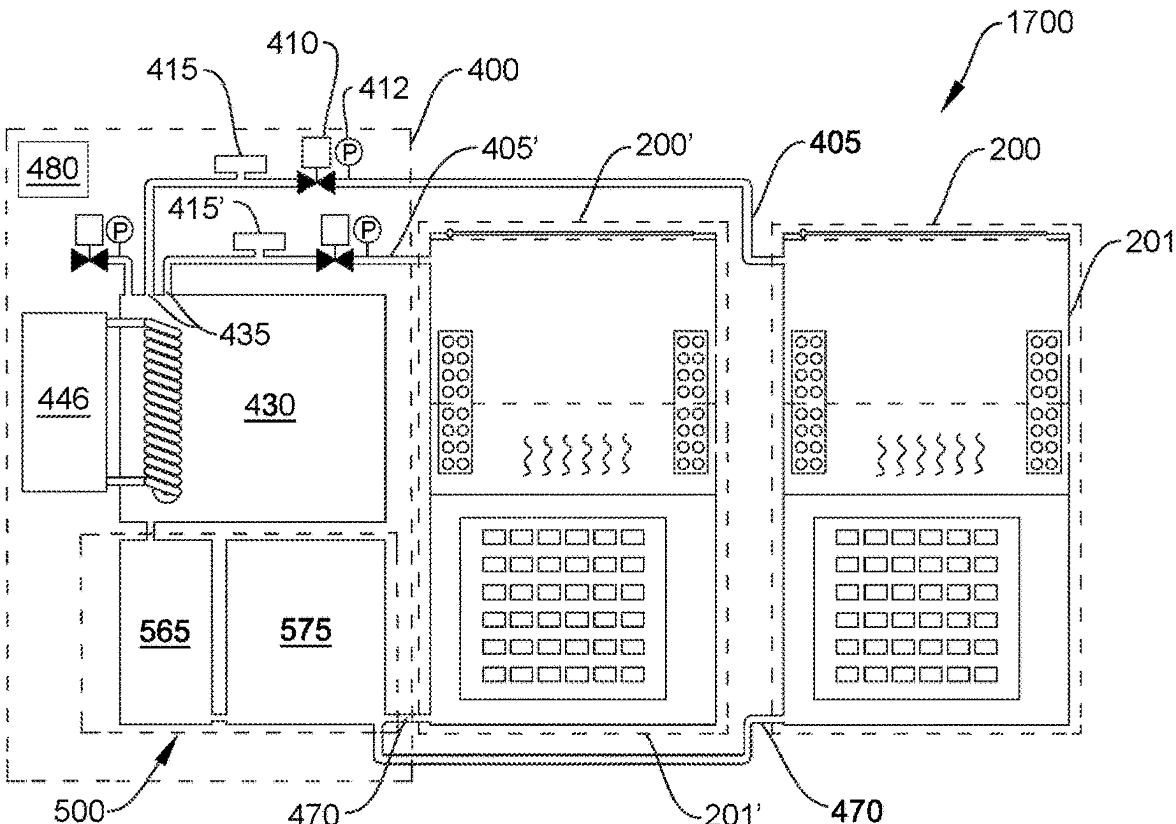


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FIG. 17

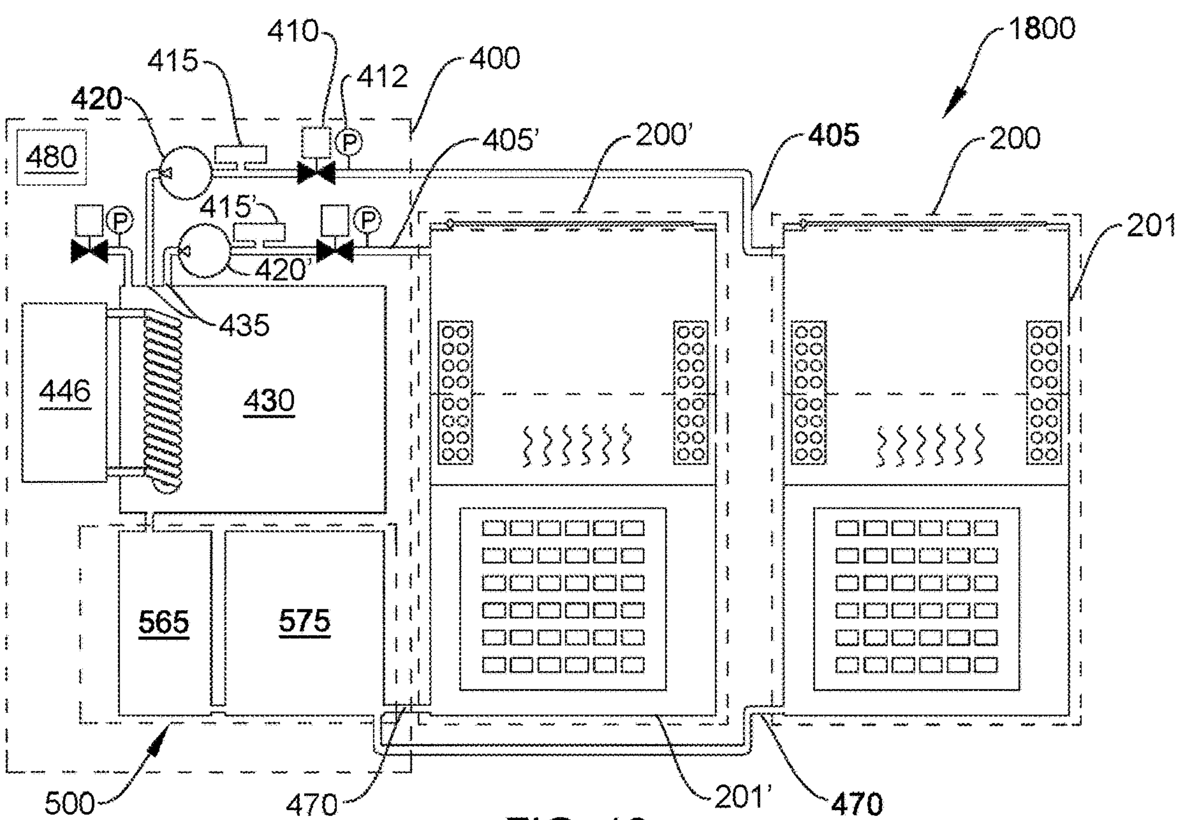

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FIG. 18

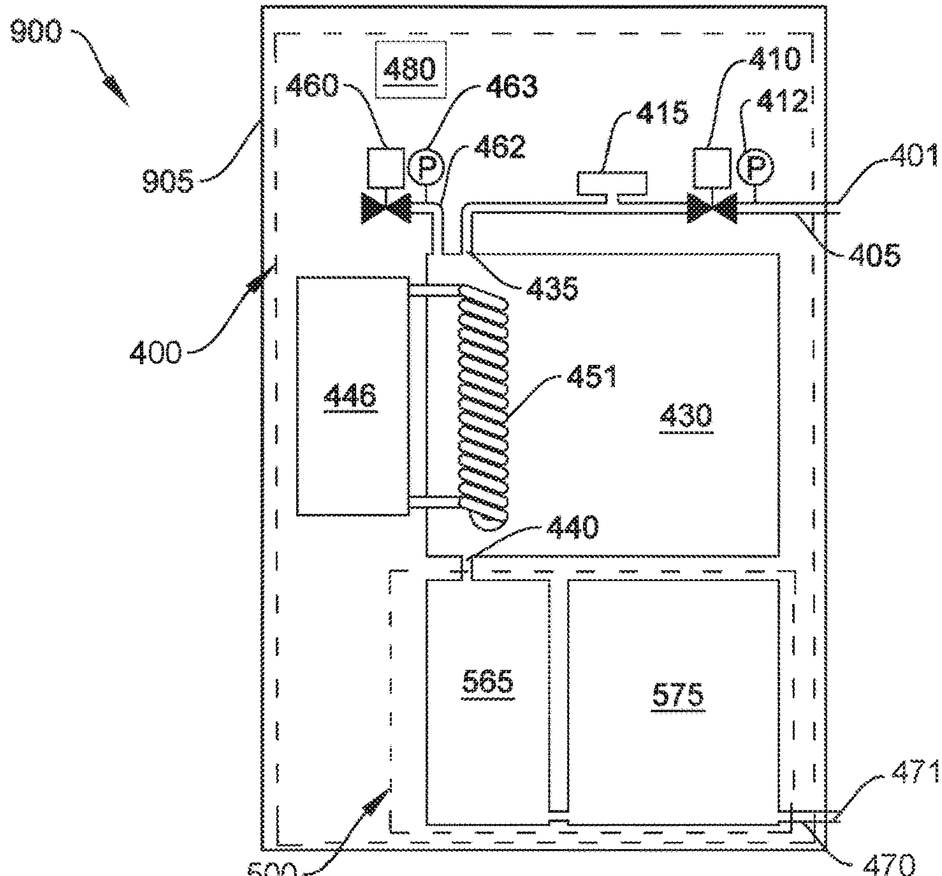
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FIG. 19

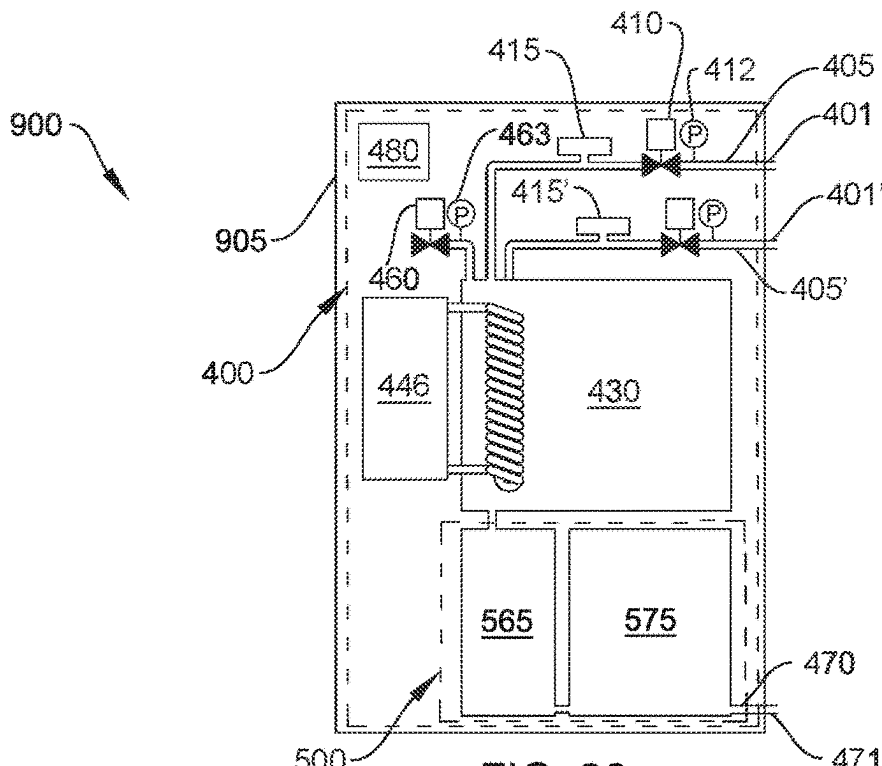
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FIG. 20

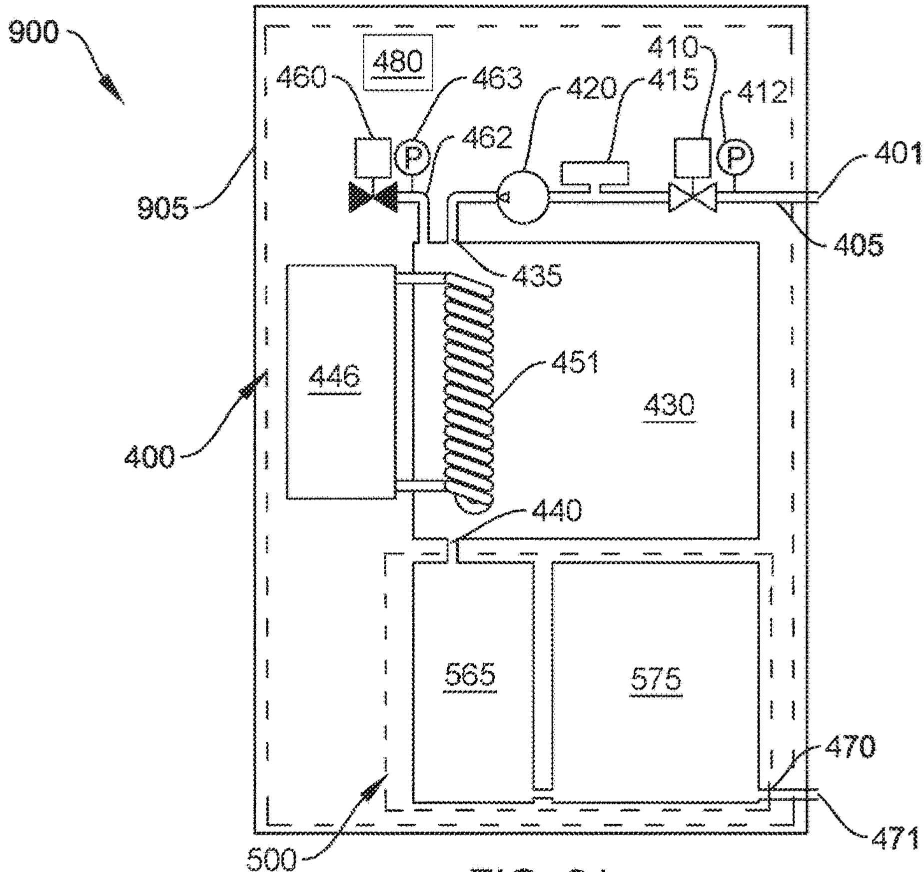
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FIG. 21

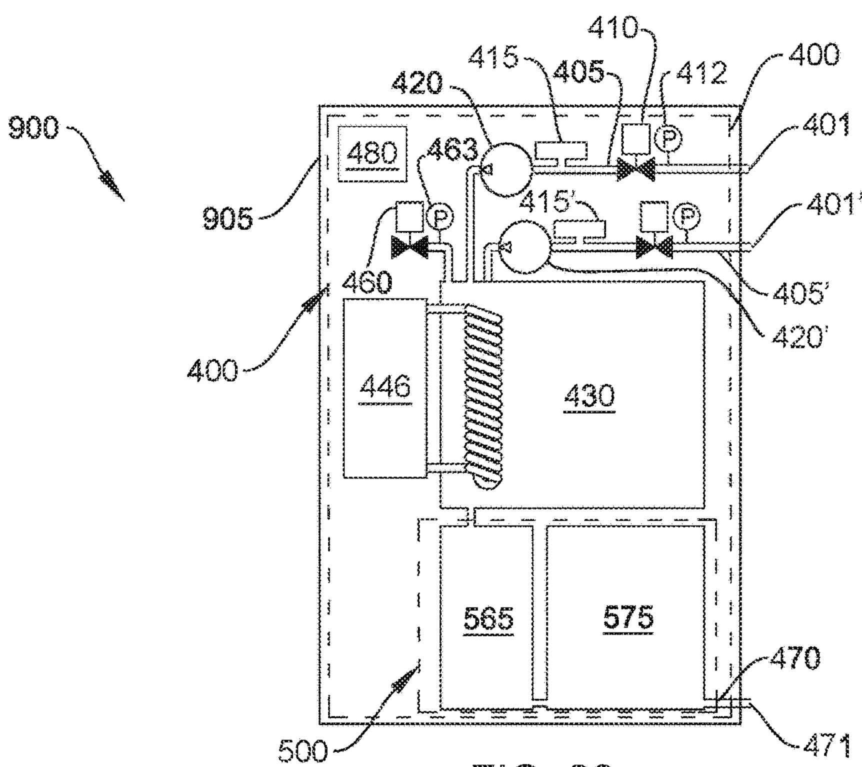
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FIG. 22

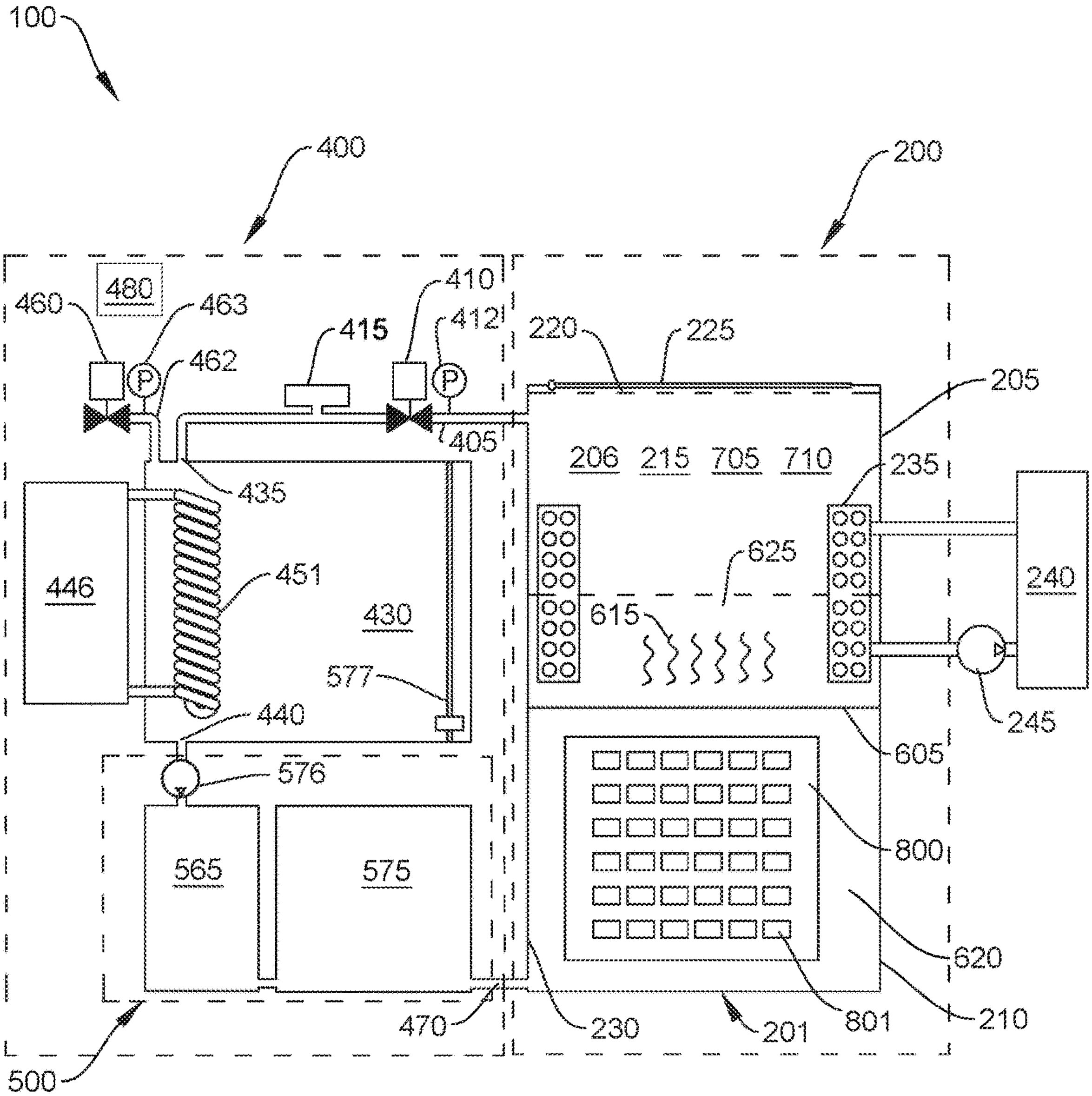
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FIG. 23

MODULAR DATA CENTER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/101,872, filed on Jan. 26, 2023, which is a continuation of International Application PCT/EP2021/055404, filed on Mar. 3, 2021, which designates the United States and claims priority to U.S. application Ser. No. 16/939,570, filed on Jul. 27, 2020, now U.S. Pat. No. 10,966,349, each of which is hereby incorporated by reference in its entirety.

FIELD

[0002] This disclosure relates to modular data centers with two-phase immersion cooling apparatuses and methods for cooling electronic devices.

BACKGROUND

[0003] Data centers house information technology (IT) equipment for the purposes of storing, processing, and disseminating data and applications. IT equipment may include electronic devices, such as servers, storage systems, power distribution units, routers, switches, and firewalls.

[0004] During use, IT equipment consumes electricity and produces heat as a byproduct. A data center containing thousands of servers requires a dedicated IT cooling system to manage the heat produced. The heat must be captured and rejected from the data center. If the heat is not removed, ambient temperature within the data center will rise above an acceptable threshold and temperature-induced performance throttling of electronic devices (e.g., microprocessors) may occur.

[0005] Data centers are energy-intensive facilities. It is not uncommon for a data center to consume over 50 times more energy per square foot than a typical commercial office building. Collectively, data centers account for about 3% of global electricity use.

[0006] Electricity use in data centers is attributable to a variety of systems, including IT equipment, air management systems, mechanical systems, electrical systems (e.g., power conditioning systems), and cooling systems for IT equipment. Examples of IT cooling systems include precision air conditioning systems, direct expansion systems, chilled water systems, free cooling systems, humidification systems, and direct liquid cooling systems. In some data centers, IT cooling and power conditioning systems account for over half of all electricity use.

[0007] Most data centers utilize precision air conditioning systems for IT cooling. Precision air conditioners employ a vapor-compression cycle, similar to residential air conditioners. Although air conditioning technology is well-suited for comfort-cooling office space, it is not well-suited for cooling thousands of relatively small, hot devices distributed throughout a large data center. Air has a relatively low heat capacity, which necessitates movement and conditioning of large amounts of air to cool IT equipment. Consequently, air conditioners suffer from poor thermodynamic efficiency, which translates to high operating expense. To reduce operating expense, there is a need to cool servers more efficiently.

SUMMARY

[0008] Advancements that improve efficiency, performance, reliability, and sustainability of data centers are needed.

[0009] In one aspect, a modular data center can include a shipping container and an immersion tank positioned within the shipping container. The immersion tank can include an upper portion, a lower portion, and a primary condenser in thermal communication with an interior volume of the immersion tank. The modular data center can include a vapor management system fluidly connected to the immersion tank. The vapor management system can include a condensing chamber having an inlet, an outlet, and an auxiliary condenser in thermal communication with an interior volume of the condensing chamber, a vapor supply passage fluidly connecting the upper portion of the immersion tank to the inlet of the condensing chamber, a valve in the vapor supply passage between the upper portion of the immersion tank and the inlet of the condensing chamber, and a liquid return passage fluidly connecting the outlet of the condensing chamber to the immersion tank. The modular data center can include an external heat rejection system positioned outside the shipping container and fluidly connected to the primary condenser. The external heat rejection system can be an evaporative cooling tower. The external heat rejection system can be a dry cooler. The modular data center can include a variable volume chamber fluidly connected to the vapor supply passage between the upper portion of the immersion tank and the inlet of the condensing chamber, wherein the condensing chamber is a fixed volume condensing chamber. The modular data center can include a pressure relief valve fluidly connected to the condensing chamber. The modular data center can include a pressure sensor located in the immersion tank or in the vapor supply passage and configured to detect pressure within the immersion tank. The modular data center can include an exhaust passage fluidly connected to the condensing chamber, a pressure relief valve in the exhaust passage, and a pressure sensor located in the vapor management system and configured to detect pressure within the vapor management system. The modular data center can include a water separator fluidly connected to the liquid return passage between the outlet of the condensing chamber and the immersion tank. The modular data center can include a liquid pump fluidly connected to the liquid return passage between the outlet of the condensing chamber and the immersion tank, a drying filter fluidly connected to the liquid return passage between the liquid pump and the immersion tank, and an impurity filter fluidly connected to the liquid return passage between the drying filter and the immersion tank.

[0010] In another aspect, a modular data center can include a shipping container and a two-phase immersion cooling apparatus positioned within the shipping container. The two-phase immersion cooling apparatus can include an immersion tank comprising an upper portion, a lower portion, and a primary condenser in thermal communication with an interior volume of the immersion tank, and a vapor management system fluidly connected to the immersion tank. The vapor management system can include a condensing chamber having an inlet, an outlet, and an auxiliary condenser in thermal communication with an interior volume of the condensing chamber, a vapor supply passage fluidly connecting the upper portion of the immersion tank to the inlet of the condensing chamber, a solenoid valve in

the vapor supply passage between the upper portion of the immersion tank and the inlet of the condensing chamber, a liquid return passage fluidly connecting the outlet of the condensing chamber to the immersion tank, a pressure sensor configured to detect pressure in the immersion tank, and an electronic control unit configured to receive a pressure signal from the pressure sensor and send a command signal to the solenoid valve. The modular data center can include an external heat rejection system positioned outside the shipping container and fluidly connected to the primary condenser. The external heat rejection system can be an evaporative cooling tower or a dry cooler. The condensing chamber can have a volume at least 10% as large as a headspace volume of the immersion tank. The vapor management system can include a chiller fluidly connected to the auxiliary condenser.

[0011] In another aspect, a method can include providing a modular data center having a shipping container, a two-phase immersion cooling apparatus positioned within the shipping container, and an external heat rejection system positioned outside the shipping container and fluidly connected to the primary condenser. The two-phase immersion cooling apparatus can include an immersion tank comprising an upper portion, a lower portion, and a primary condenser in thermal communication with an interior volume of the immersion tank, and a vapor management system. The vapor management system can include a condensing chamber having an inlet, an outlet, and an auxiliary condenser in thermal communication with an interior volume of the condensing chamber, a vapor supply passage fluidly connecting the upper portion of the immersion tank to the inlet of the condensing chamber, a valve in the vapor supply passage between the upper portion of the immersion tank and the inlet of the condensing chamber, and a liquid return passage fluidly connecting the outlet of the condensing chamber to the immersion tank. The method can include detecting a pressure within the immersion tank. The method can include opening the valve when the pressure is greater than a predetermined threshold setting and admitting dielectric vapor from the immersion tank into the condensing chamber. The method can include closing the valve when the pressure in the immersion tank is less than the predetermined threshold setting. The method can include condensing the dielectric vapor to a liquid in the condensing chamber. The method can include returning the liquid to the immersion tank through the liquid return passage. The method can include circulating a coolant through the primary condenser and external heat rejection system where the coolant has a temperature greater than or equal to an ambient air temperature. The method can include circulating a first coolant from the primary condenser to the external heat rejection system and circulating a second coolant through the auxiliary condenser, where a temperature of the first coolant is greater than a temperature of the second coolant. The method can include providing a vapor pump fluidly connected to the vapor supply passage between the upper portion of the immersion tank and the inlet of the condensing chamber and operating the vapor pump while the valve is open to purge gas from a headspace of the immersion tank and reduce the pressure within the immersion tank below atmospheric pressure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. **1** shows a perspective view of a modular data center, in accordance with some embodiments of the present invention.

[0013] FIG. **2** shows a partial cutaway view of the modular data center of FIG. **1** exposing a plurality of immersion cooling tank assemblies within a container, in accordance with some embodiments of the present invention.

[0014] FIG. **3** shows a perspective view of an immersion cooling tank assembly, in accordance with some embodiments of the present invention.

[0015] FIG. **4** shows a perspective view of a plurality of immersion cooling tank assemblies positioned in a conventional data center, in accordance with some embodiments of the present invention.

[0016] FIG. **5** shows a schematic diagram of a two-phase immersion cooling apparatus with a vapor management system, in accordance with some embodiments of the present invention.

[0017] FIG. **6** shows the apparatus of FIG. **5** with an open flow control valve, in accordance with some embodiments of the present invention.

[0018] FIG. **7** shows the apparatus of FIG. **5** with an open flow control valve and expanded bellows, in accordance with some embodiments of the present invention.

[0019] FIG. **8** shows a plot of immersion tank pressure and electronic device power consumption versus time, in accordance with some embodiments of the present invention.

[0020] FIG. **9** shows the apparatus of FIG. **5** with an open pressure relief valve, in accordance with some embodiments of the present invention.

[0021] FIG. **10** shows a gravity-based water separator and filtration assembly, in accordance with some embodiments of the present invention.

[0022] FIG. **11** shows an alternate embodiment of a water separator, in accordance with some embodiments of the present invention.

[0023] FIG. **12** shows a chiller and auxiliary condenser, in accordance with some embodiments of the present invention.

[0024] FIG. **13** shows the apparatus of FIG. **5** with a vapor pump included in the vapor management system, in accordance with some embodiments of the present invention.

[0025] FIG. **14** shows a plot of immersion tank pressure and electronic device power consumption versus time when tank pressure is drawn below atmospheric pressure prior to powering on the electronic device, in accordance with some embodiments of the present invention.

[0026] FIG. **15** shows a prior art immersion cooling system with a primary condenser.

[0027] FIG. **16** shows a prior art immersion cooling system with a primary condenser and a freeboard condenser.

[0028] FIG. **17** shows an embodiment of a two-phase immersion cooling apparatus with two immersion tanks fluidically connected to a central vapor management system, in accordance with some embodiments of the present invention.

[0029] FIG. **18** shows the apparatus of FIG. **17** with a vapor pump fluidically connected between each immersion tank and the central vapor management system, in accordance with some embodiments of the present invention.

[0030] FIG. **19** shows a vapor processing apparatus, in accordance with some embodiments of the present invention.

[0031] FIG. **20** shows a vapor processing apparatus with two vapor supply inlets, in accordance with some embodiments of the present invention.

[0032] FIG. **21** shows a vapor processing apparatus with a vapor pump, in accordance with some embodiments of the present invention.

[0033] FIG. **22** shows a vapor processing apparatus with two vapor supply inlets and two vapor pumps, in accordance with some embodiments of the present invention.

[0034] FIG. **23** shows the apparatus of FIG. **5** with a liquid level sensor included in the vapor management system and a liquid pump included in the water separator and filtration assembly, in accordance with some embodiments of the present invention.

DETAILED DESCRIPTION

[0035] Direct liquid cooling systems present a promising alternative to air conditioning systems for data center applications. One form of direct liquid cooling is immersion cooling. In an immersion cooling system, an electronic device is immersed in dielectric fluid. Waste heat from the electronic device is transferred to the fluid and then rejected outside the data center. Since waste heat is not released into the ambient air of the data center, a precision air conditioning system is generally not needed.

[0036] Immersion cooling systems may employ single-phase or two-phase technology. In a single-phase immersion cooling system, electronic devices are immersed in a fluid, such as mineral oil. Waste heat from the electronic devices is transferred to and warms the fluid. The warmed fluid is pumped from the immersion cooling system to a heat rejection system, such as an evaporative cooling tower, dry cooler, or chilled water loop, that captures waste heat from the fluid and rejects the heat outside the data center.

[0037] A drawback of single-phase immersion cooling technology is that mineral oil can act as a solvent and, over time, remove identifying information from motherboards, processors, and other components. For instance, product labels (e.g., stickers containing serial numbers and bar codes) and other markings (e.g., screen printed information and model numbers on capacitors and other devices) may dissolve and wash off due to a continuous flow of mineral oil over device surfaces. As the labels and ink wash off the server, the mineral oil can become contaminated and may need to be replaced, resulting in expense and periodic downtime. Another drawback of single-phase immersion cooling is that servers cannot be serviced immediately after being withdrawn from the tank. Typically, the server must be removed from the tank and allowed to drip dry for several hours before servicing. During this drying period, the server may be exposed to contaminants in the circulating air, and the presence of mineral oil on the server may attract and trap contaminants (e.g., dust or particulates) on sensitive circuitry, which may increase risk of short-circuiting and failure.

[0038] In a two-phase immersion cooling system, an electronic device is immersed in a fluid, such as hydrofluoroether, in a tank. The fluid readily evaporates and leaves no residue, so two-phase systems do not suffer the drawbacks of single-phase mineral oil-based systems mentioned above. During use, waste heat from the electronic device is absorbed by the fluid, resulting in localized vaporization of the fluid. Vapor rises into a headspace of the tank and is condensed by a condenser. Heat from the vapor is transferred to coolant circulating through the condenser, thereby warming the coolant. The warmed coolant is then pumped from the condenser to a heat rejection system, such as an evaporative cooling tower, dry cooler, or chilled water loop, which captures the waste heat from the fluid and rejects the heat outside the data center.

[0039] Two-phase immersion cooling systems leverage the benefits of phase change heat transfer, which makes them more efficient at and capable of cooling electronic devices with high heat flux, such as high-performance computing (HPC) servers with multiple graphics processing units (GPUs), than single-phase immersion cooling systems. However, along with the benefits of phase change heat transfer comes a challenge. In practice, retaining vapor within the system has proven challenging in prior art two-phase systems. Over time, existing two-phase immersion cooling systems inevitably suffer fluid loss to the environment. The lost fluid may be costly to replenish on a recurring basis. Examples of prior art two-phase cooling systems and their modes of fluid loss and other shortcomings are described below.

[0040] FIG. **15** shows a prior art example of a basic two-phase immersion cooling apparatus **1500**. The apparatus **1500** includes an immersion tank **201** partially filled with dielectric fluid **620** in liquid phase. The apparatus includes a condenser **235** mounted in a headspace of the tank **201**. An electronic device **800** is immersed in the dielectric fluid. The electronic device may be a server including one or more microprocessors **801**. The tank **201** is enclosed by a lid **225**. When powered on and functioning, the electronic device **800** produces heat. The heat is transferred to the dielectric fluid **620**, which causes a portion of the fluid to boil and dielectric vapor **615** to form. The vapor **615** rises through a bath of dielectric liquid **620** and enters the headspace of the tank **201**. When vapor **615** contacts the condenser **235**, it condenses to liquid and passively recirculates back to the liquid bath, thereby completing a cycle of evaporation, condensation, precipitation, and collection. During operation, boiling of the relatively dense dielectric fluid **620** produces a relatively less dense vapor **615**, which expands and enters the headspace occupied by non-condensable gases. Dielectric fluid **620** occupies more volume as a vapor than liquid, so as more vapor **615** is produced and enters the headspace, tank pressure increases. To prevent the tank pressure from reaching an unsafe level, a pressure relief valve **460** is provided in the tank **201** and opened when the pressure exceeds a predetermined threshold. Upon actuation of the pressure relief valve **460**, dielectric vapor **615** is released from the tank **201** and lost to the environment. Over time, periodic valve actuation and fluid loss depletes the fluid **620**, necessitating replenishment.

[0041] In a variation of the example shown in FIG. **15**, the two-phase immersion cooling apparatus **1500** can be configured to maintain a tank pressure near ambient pressure (e.g., 1 atm) at all times. Operating at or near atmospheric pressure may be desirable to minimize fluid losses caused by leakage or diffusion through system joints or materials, respectively. When heat load from the electronic device **800** increases, the rate of dielectric vapor production increases and pressure builds within the tank **201**. To avoid pressure accumulation, the pressure relief valve **460** may be actuated whenever tank pressure rises above atmospheric pressure. During periods of intense computing, this may result in frequent venting of vapor **615** to the environment. During idle periods, the heat load from the device **800** will decrease, and the rate of vapor production will decrease or even cease. Any remaining vapor **615** is condensed by the condenser

235 and, subsequently, air in the headspace is further cooled by the condenser **235**, resulting in the tank pressure dropping below ambient pressure. To alleviate negative pressure in the tank **201**, the pressure relief valve **460** may open and allow ambient air to enter the tank **201**. Over time, this cyclical gas exchange results in fluid loss.

[0042] In another variation of the example shown in FIG. **15**, the two-phase immersion cooling apparatus **1500** can be equipped with a high-capacity condenser **235** having a cooling capacity that exceeds a maximum heat load of the electronic device **800**. The condenser **235** may operate at a temperature significantly below the vapor temperature of the dielectric fluid **620**, thereby ensuring that vapor **615** is promptly condensed, thereby avoiding over-pressurization due to vapor accumulation. Although this approach reduces fluid loss, the condenser **235** is large and energy-inefficient, making it costly and impractical for large-scale data center applications.

[0043] In another variation of the example shown in FIG. **15**, the two-phase immersion cooling apparatus **1500** can be hermetically sealed. The tank **201** can be a pressure vessel, capable of withstanding high positive or negative pressures without risk of failure. Unfortunately, pressure vessels are costly to construct and maintain. For the sake of product liability, the effect of high-pressure operation on unique server models would require evaluation and approval before use, resulting in ongoing validation and liability burdens. High operating pressures may promote diffusion losses of fluid and accelerate aging of gaskets and other system components. In a sealed system, routine server maintenance requires shutting down the cooling system and unsealing the lid to access to the servers, which is time-consuming. For these reasons, a hermetically-sealed tank **201** is not a practical option for most data centers, especially those with high up-time requirements.

[0044] FIG. **16** shows a second prior art example of a two-phase immersion cooling apparatus **1600**. The apparatus **1600** includes an immersion tank **201** partially filled with dielectric fluid **620** in liquid phase. An electronic device **800** is immersed in the dielectric fluid **620**. The electronic device **800** may be a server including one or more microprocessors **801**. The immersion tank **201** is enclosed by a lid **225**. The apparatus **1600** can include two condensers. For example, the apparatus **1600** may include a primary condenser **235** and a freeboard condenser **250** mounted within the immersion tank **201**. The primary condenser **235** may be located above a liquid line **605** in the headspace of the tank **201**. The freeboard condenser **250** may be located a distance above the primary condenser **235** in the headspace **206**. In one example, the primary condenser **235** may operate at a temperature of about 5° C. to 15° C. The freeboard condenser **250** may operate at a lower temperature of about −28° C. to −2° C. The apparatus **1600** has a high freeboard ratio, where freeboard ratio is defined as a distance measured from the top of the primary condenser **235** to an underside of the lid **225** divided by an internal width of the immersion tank **201**.

[0045] During steady-state operation of the apparatus **1600**, vapor **615** is generated as heat from the electronic device **800** vaporizes fluid **620** in the tank **201**. The vapor **615** is heavier than air **705**, so a first zone **1605** containing saturated vapor **615** may settle above the liquid line **605**. A second zone **1610** containing mixed vapor **615** and air **705** may form above the saturated vapor **615**. A third zone **1615** containing mostly air **705** may form above the mixture of vapor **615** and air **705**. The saturated vapor zone **1605** may be located between the liquid line **605** and the primary condenser **235**. The mixed vapor and air zone **1610** may be located between the primary condenser **235** and the freeboard condenser **250**. The third zone **1615** containing mostly air **705** may be located between the freeboard condenser **250** and the lid **225**. The primary condenser **235** may be appropriately sized to condense most of the vapor **615** produced during steady-state operation. The freeboard condenser **250** may condense vapor **615** that rises above the primary condenser **235** and enters the second zone **1610**. During steady-state operation, an equilibrium of vapor production and condensing may exist.

[0046] During periods of high microprocessor **801** utilization, more electric power is consumed by the device **800** and more heat is produced, resulting in a higher rate of vapor production. As the amount of vapor **615** in the headspace **206** increases, the depth of the saturated vapor zone **1605** grows. The freeboard condenser **250**, which is maintained at a much lower temperature than the primary condenser **235**, may effectively condense vapor **615** that reaches it.

[0047] Although effective, the apparatus **1600** in FIG. **16** has certain drawbacks. First, the apparatus **1600** is inefficient, since the freeboard condenser **250** requires a chiller to operate continuously to maintain a suitably cold temperature. Second, the apparatus **1600** is not compact or user-friendly. To be effective, the apparatus **1600** must have a high freeboard ratio, which necessitates a relatively tall tank **201**. While a tall tank **201** may be acceptable in a conventional data center with a high ceiling and access to a hoist or ladder to insert and remove electronic equipment to and from the tank **201**, it is not suitable for a compact application, such as an edge or mobile data center application, where the system **1600** is installed in a confined space (e.g., shipping container **1005** (FIGS. **1** and **2**) or utility enclosure) with height restrictions and minimal overhead clearance. In addition to these practical limitations, the cooling system **1600** in FIG. **16** may also suffer from fluid loss through similar modes as the cooling system **1500** shown in FIG. **15** and described above.

[0048] In view of the foregoing examples, it is desirable to provide a two-phase immersion cooling apparatus that is compact, energy-efficient, inexpensive, and experiences minimal fluid loss to the environment.

Two-Phase Immersion Cooling Apparatus with Active Vapor Management

[0049] FIG. **5** shows an embodiment of a two-phase immersion cooling apparatus **100** with active vapor management in accordance with certain embodiments of the invention. The two-phase immersion cooling apparatus **100** may be used in a variety of applications, ranging from a modular data center **1000**, as shown in FIG. **1**, to a traditional data center **2000**, as shown in FIG. **4**. In the embodiment of FIG. **1**, the immersion cooling apparatus **100** may be positioned within a container **1005** (FIGS. **1** and **2**) and fluidically connected to an external heat rejection system **240** mounted atop the container **1005**.

[0050] The apparatus **100** may include an immersion tank assembly **200**. The immersion tank assembly **200** may include an immersion tank **201** that is selectively sealable with a lid **225** (FIG. **3**). The immersion tank **201** may be vertically compact, allowing it to be placed in confined spaces, such as shipping containers **1005** or utility enclosures associated with modular or edge data center applications. The immersion tank **201** may have a height that is less than a length or width of the tank **201**. The immersion tank **201** may have a height that is less than a length and less than a width of the tank **201**.

[0051] The immersion tank **201** may have an upper portion **205** and a lower portion **210**. The upper portion **205** may be a portion of the immersion tank **201** that is located above the liquid line **605**. The lower portion **210** may be a portion of the immersion tank **201** that is located below the liquid line **605**. The liquid line **605** may be an interface formed between gases (e.g., air and dielectric vapor) in the headspace and dielectric liquid **620** in the lower portion **210** of the immersion tank **201**. The immersion tank **201** may have an opening **220** in the upper portion **205**. The tank **201** may have an electrical insulating layer **230** on an interior surface. The immersion tank may have a lid **225**. When open, the lid **225** may provide access to an interior volume **215** of the immersion tank **201** to facilitate insertion and removal of electronic devices **800** (e.g., servers, switches, or power electronics). When closed, the lid **225** may enclose the opening **220** and prevent vapor loss. The lid **225** may (e.g., hermetically-) seal the opening **220**.

[0052] The immersion tank **201** may be partially filled with (e.g., dielectric) fluid **620**. The fluid **620** may be selected, or formulated by mixing two or more fluids, to have a boiling point that is less than an operating temperature of a heat-generating electronic device **800**, such as a microprocessor **801**. When the electronic device **800** is operating, fluid **620** in contact with the device **800** may boil locally and produce vapor **615**. Vapor **615** may rise through the fluid bath and into the headspace **206** of the immersion tank **201**. The vapor **615** may settle atop the liquid line **605**, forming a blanket of saturated vapor **625**.

[0053] The immersion tank assembly **200** may include a primary condenser **235**. The primary condenser **235** may be located in the headspace **206** of the tank **201**. The primary condenser **235** may condense vapor **615** within the immersion tank **201**. In some implementations, the primary condenser **235** may be a cooling coil and, more specifically, a cooling coil that receives coolant, such as chilled water, water-glycol mixture, refrigerant, and the like from a heat rejection system **240**, such as an evaporative cooling tower, dry cooler, or chilled water loop. The heat rejection system **240** may include a coolant pump **245**, as shown in FIGS. **1** and **5**. The coolant pump **245** may circulate coolant through the primary condenser **235** and heat rejection system **240**.

[0054] To minimize energy consumption, the primary condenser **235** may operate at a temperature at or slightly above room temperature. In one embodiment, the primary condenser **235** may receive and circulate coolant at a temperature of about 33° C. when ambient temperature is 30° C. In another embodiment, the primary condenser **235** may receive and circulate coolant at a temperature of about 25° C. to 40° C. In yet another embodiment, the primary condenser **235** may receive and circulate coolant at a temperature of about 30° C. to 36° C. In still another embodiment, the primary condenser **235** may receive and circulate coolant at a temperature of about 0 to 10 degrees above an ambient air temperature. In still another embodiment, the primary condenser **235** may receive and circulate coolant at a temperature of about 0 to 15 degrees above an ambient air temperature.

[0055] The apparatus **100** may include a vapor management system **400**. The vapor management system **400** may be fluidically connected to the immersion tank **201**. The vapor management system **400** may receive vapor **615** from the immersion tank **201** when necessary to avoid overpressurization of the tank **201**, condense the vapor **615** to liquid **620**, and return the liquid **620** to the immersion tank **201** for reuse. The vapor management system **400** may be located at least partially outside the headspace **206** of the immersion tank **201**.

[0056] The vapor management system **400** may be an auxiliary vapor management system, for example, an external vapor management system. The vapor management system **400** may activate during periods of high, variable, or sustained vapor production. The vapor management system **400** may be activated or deactivated based on one or more system variables (e.g., tank pressure, tank temperature, or device power). The vapor management system **400** may provide surplus condensing capacity to manage periods of increased heat load and vapor production, thereby supplementing the condensing capacity of the primary condenser **235** when needed.

[0057] The vapor management system **400** may be actively controlled based on conditions measured or determined in the apparatus **100**. For embodiment, the vapor management system **400** may be controlled based on an input of one or more variables, such as pressure, temperature, device power, vapor concentration, or opacity within the immersion tank **201**. A variable may be measured with an electronic sensor, mechanically detected, estimated based on a correlated variable, or determined through any other suitable technique and combinations thereof.

[0058] The vapor management system **400** may include a vapor supply passage **405**. The vapor supply passage **405** may fluidically connect the vapor management system **400** to the upper portion **205** of the immersion tank **201**. The vapor supply passage **405** may be any suitable type of fluid passage, such as, for example, a tube, pipe, integrally-formed passage, or combinations thereof.

[0059] The vapor management system **400** may include a liquid return passage **470**. The liquid return passage **470** may fluidically connect the vapor management system **400** to the lower portion **210** of the immersion tank **201**. The liquid return passage **470** may be any suitable type of fluid passage, such as a tube, pipe, or integrally-formed passage, and combinations thereof. Together, the vapor supply passage **405** and liquid return passage **470** may enable circulation of fluid from and to the immersion tank **201**. For example, the vapor management system **400** may receive dielectric vapor **615** from the immersion tank **201** and return liquid dielectric fluid **620** to the immersion tank **201**.

[0060] The vapor management system **400** may include a valve **410** in the vapor supply passage **405**. The (e.g., flow control) valve **410** may control flow of vapor through the vapor supply passage **405**. When open, the valve **410** may permit vapor flow through the vapor supply passage **405** and from the immersion tank **201** to the vapor management system **400**. The valve **410** may be a manual or automatic valve. The valve **410** may have a threshold (e.g., fixed or variable) pressure setting. In one embodiment, the threshold pressure setting may be about 0.15 psig. In this embodiment, when pressure in the immersion tank is greater than or equal to 0.15 psig, the valve **410** will open. The valve **410** may remain open until the vapor pressure in the immersion tank

201 drops below 0.15 psig, at which time the valve **410** may close. In another embodiment, the threshold pressure setting may be at or between −0.15 psig and 0.15 psig. In another embodiment, the threshold pressure setting may be at or between −0.25 psig and 0.25 psig. In another embodiment, the threshold pressure setting may be at or between −0.9 psig and 0.9 psig. In another embodiment, the threshold pressure setting may be at or between 0 psig and 0.25 psig. In another embodiment, the threshold pressure setting may be at or between −0.25 psig and 0 psig. In another embodiment, the threshold pressure setting may be at or between 1 psig and 5 psig. In another embodiment, the threshold pressure setting may be at or between 4 psig and 10 psig. In another embodiment, the threshold pressure setting may be at or between −1 psig and −5 psig. In another embodiment, the threshold pressure setting may be at or between −4 psig and −10 psig.

[0061] In some embodiments, the threshold pressure setting may be variable instead of fixed. A variable pressure setting may be useful in addressing an anticipated surge in vapor production during transient operation. For example, when device power consumption increases rapidly, a time lag may occur before a rise in vapor pressure is detected through pressure monitoring. The rise in vapor production may be accurately predicted by monitoring device power consumption. Upon detecting an increase in device power consumption. the threshold setting may be temporarily reduced to activate the vapor management system **400** sooner than if a fixed threshold setting were used.

[0062] In some applications, the vapor management system **400** may include a condensing chamber **430**. The condensing chamber may be a fixed volume condensing chamber. The condensing chamber **430** may have an interior volume disposed between an inlet **435** and an outlet **440**. The vapor supply passage **405** may fluidically connect to the inlet **435** of the condensing chamber **430**. The vapor supply passage **405** may fluidically connect an outlet of the valve **410** to the inlet **435** of the condensing chamber **430**. The vapor supply passage **405** may transport vapor **615** from the immersion tank **201** to the condensing chamber **430** when the valve **410** is open. The condensing chamber **430** may have a volume at least 10% as large as the headspace volume of the immersion tank **201**. The condensing chamber **430** may have a volume at least 30% as large as the headspace volume of the immersion tank **201**. The condensing chamber **430** may have a volume that is at least 50% as large as the headspace volume of the immersion tank **201**. The condensing chamber **430** may have a volume that is at least 70% as large as the headspace volume of the immersion tank **201**. Headspace volume may be a volume measured between the liquid line **605** and interior surface of the lid **225** and bounded by the sidewalls of the immersion tank **201**.

[0063] The condensing chamber **430** may include an auxiliary condenser **451**. The auxiliary condenser **451** may be in thermal communication with the condensing chamber **430**. The auxiliary condenser **451** may extract heat from vapor **615** and condense the vapor **615** in the condensing chamber **430**. The auxiliary condenser **451** may extend into an interior volume of the condensing chamber **430** or be in contact with at least one surface of the condensing chamber **430**. The auxiliary condenser **451** may operate at a lower temperature than the primary condenser **235**. In one embodiment, the auxiliary condenser **451** may include a cooling coil connected to a liquid chiller **446**. The liquid chiller **446** may circulate chilled coolant through the auxiliary condenser **451**. In one embodiment, the liquid chiller **446** may circulate liquid at a temperature of about 5 to 15 degrees C. In another embodiment, the liquid chiller **446** may circulate liquid at a temperature of about-2 to 10 degrees C. In another embodiment, the liquid chiller **446** may circulate liquid at a temperature of about −10 to −5 degrees C. In another embodiment, the liquid chiller **446** may circulate liquid at a temperature of about −28 to −2 degrees C.

[0064] In one variation, the liquid chiller **446** may be a refrigeration system, as shown in FIG. **12**. The chiller **446** may include a pump **452** that circulates fluid between a reservoir **453** and a cooling coil **451** to maintain the cooling coil at a temperature (Tc) that is less than the ambient temperature. The fluid may be a dielectric fluid. The fluid may be the same type of fluid used in the immersion tank **201** to reduce risk of cross-contamination from leakage or diffusion. Using the same type of fluid may also simplify maintenance tasks. The liquid chiller **446** may include a compressor **449**, a condenser **447**, an expansion valve **450**, and an evaporator **448**. The liquid chiller **446** may employ a refrigeration cycle to extract heat from the fluid in the reservoir **453** and reject heat through the condenser **447**.

[0065] The vapor management system **400** may include a variable volume chamber **415** (e.g., a bellows). The variable volume chamber **415** may be made of a vapor-resistant material, such as a metalized polyester film (e.g., mylar), with an expandable interior volume. The variable volume chamber **415** may be deflated when the pressure in the vapor management system **400** is less than or equal to 1 atm. The variable volume chamber **415** may inflate when pressure within the vapor management system **400** is greater than 1 atm. The variable volume chamber **415** may allow the total volume of the vapor management system **400** to expand to increase total vapor capacity, thereby allowing the system **400** to receive more vapor **615** during transient periods. Expansion of the variable volume chamber **415** may reduce pressure of incoming vapor **615** and promote condensing of the vapor **615**.

[0066] The vapor management system **400** may include a pressure relief valve **460**. The pressure relief valve **460** may be a safety device. The pressure relief valve **460** may open at a predetermined pressure threshold to prevent overpressurization of the vapor management system **400**. The pressure relief valve **460** may be fluidically connected to the condensing chamber **430** via an exhaust passage **462**. In one embodiment, the pressure relief valve **460** may be configured to open when pressure within the vapor management system **400** is equal to or greater than about 0.15 psig. In another embodiment, the pressure relief valve **460** may be configured to open when pressure within the vapor management system **400** is equal to or greater than about 0.20 psig. In yet another embodiment, the pressure relief valve **460** may be configured to open when pressure within the vapor management system **400** is equal to or greater than about 0.25 psig.

[0067] The vapor management system **400** may include a vapor pump **420**, as shown in FIG. **13**. The vapor pump **420** may be configured to purge a mixture of air and dielectric vapor **615** from the immersion tank **201** to the condensing chamber **430**. The vapor pump **420** may be located upstream of the condensing chamber **430**. The vapor pump **420** may have an inlet and an outlet. The inlet of the vapor pump **420** may be fluidically connected to the outlet of the variable volume chamber **415**. The outlet of the vapor pump **420** may be fluidically connected to the inlet **435** of the condensing chamber **430**. The vapor pump **420** may be capable of reducing the pressure in the immersion tank **201** below atmospheric pressure. The vapor pump **420** may overcome gravitational effects on fluid flow and allow the vapor management system **400** to be positioned irrespective of height or orientation relative to the headspace **206**, thereby providing greater design freedom that may be needed when packaging the vapor management system **400** in a confined space, such as a container **1005** (FIGS. **1** and **2**) or utility box.

[0068] The vapor pump **420** may be useful for preemptively addressing periods of anticipated elevated vapor production. In one embodiment, upon detecting an increase in device power consumption that will result in elevated vapor production, the valve **410** may be opened and the vapor pump **420** may be activated to purge vapor **615** from the headspace **206**, thereby drawing down vapor pressure within the tank **201** ahead of an anticipated pressure rise. This approach may diminish a pressure rise rate within the immersion tank **201** resulting from increased power consumption and vapor production.

[0069] The vapor management system **400** may be electronically controlled. The vapor management system **400** may include an electronic control unit **480**. The electronic control unit **480** can be configured to open or close the flow control valve **410** based on a signal received from a sensor **412** or other input signal. For example, the electronic control unit **480** may receive an input signal (e.g., a pressure signal) from the sensor **412** and, based on the input signal, send a command signal to the flow control (e.g., solenoid) valve **410** to open or close. The sensor **412** can be a pressure sensor configured to measure pressure in the immersion tank **201**. The sensor **412** can be configured to transmit a signal to the electronic control unit **480** corresponding to pressure in the immersion tank **201**. The signal can be transmitted through a wired or wireless connection.

[0070] The electronic control unit **480** can be configured to open or close the pressure relief valve **460** based on a signal received from a sensor or other input. The sensor **463** can be a pressure sensor configured to measure pressure in the vapor management system **400**. The sensor **463** can be configured to transmit a signal to the electronic control unit **480** corresponding to pressure in the vapor management system **400**. The signal can be transmitted through a wired or wireless connection.

[0071] The electronic control unit **480** can be configured to activate or deactivate the vapor pump **420** (FIG. **13**) based on a signal received from the (e.g., pressure) sensor **412**. For instance, in the embodiment plotted in FIG. **14**, the electronic control unit **480** may activate the vapor pump **420** and purge vapor **615** from the headspace **206** to reduce tank pressure below atmospheric pressure prior to initiating a cooling cycle. Reducing pressure within the immersion tank **201** may reduce the boiling temperature of the fluid **620**, which may be desirable in certain applications. In the embodiment of FIG. **14**, tank pressure is drawn down below atmospheric pressure and the threshold pressure setting of the valve **410** is set above atmospheric pressure, thereby allowing the operating pressure of the tank **201** to oscillate in a range that includes values above and below atmospheric pressure. This method may minimize the extent and duration of tank pressure deviations away from atmospheric pressure, thereby minimizing fluid loss due to pressure-induced diffusion or leakage. In another embodiment, tank pressure may be drawn down below atmospheric pressure and the threshold pressure setting of the valve **410** may be set below atmospheric pressure, thereby allowing the operating pressure of the tank **201** to oscillate in a negative pressure range. This method may be desirable to reduce the boiling temperature of the fluid **620**.

[0072] In some embodiments, the apparatus **100** may include a water separation and filtration system **500** (FIG. **5**). As shown in FIGS. **5**, **10**, and **11**, the water separation and filtration system **500** may include an assembly of one or more of the following components: a water separator **565** and a filtration system **575** that may include—for the purpose of illustration rather than limitation—a liquid pump **585**, a drying filter **590**, an impurity filter **580**, and the like.

[0073] For example, referring to FIG. **10**, in some embodiments, the apparatus **100** may include a water separator **565**. The water separator **565** may include an inlet **440** and an outlet **574**. The water separator **565** may receive condensed liquid from the condensing chamber **430**. The water separator **565** may be configured to separate water **715** or other undesirable fluids from dielectric fluid **620**. The water separator **565** may capture water **715** or other undesirable fluids and allow dielectric fluid **620** to pass through. Water **715** or other undesirable fluids accumulated in the water separator **565** may be drained periodically. The amount of accumulated water **715** or other undesirable fluids may depend on ambient air humidity, how well the apparatus is sealed, and how frequently the lid **225** of the immersion tank **201** is opened.

[0074] In one embodiment, the water separator **565** may be a gravity-based water separator. Water **715** or other undesirable fluids may be less dense than the dielectric liquid **620**. Consequently, captured water **715** or other undesirable fluids may settle atop the dielectric liquid **620** within the water separator **565**. The water **715** or other undesirable fluids may be purged from the water separator **565** periodically through a drain valve **573**. Dewatered dielectric liquid **620** may occupy the lower portion of the water separator **565**. Dewatered dielectric liquid **620** may be drawn from the water separator **565** through the outlet **574** located in the lower portion of the water separator **565**.

[0075] In an alternate embodiment, the water separator **565** may be a pump-based water separator. As shown in FIG. **23**, the auxiliary condenser **451** may extract heat from vapor **615** and water vapor **710** and condense the vapor **615** and water vapor **710** in the condensing chamber **430**. A liquid pump **576** may be fluidically coupled between the condensing chamber **430** and the water separator **565**. The condensate resulted from condensing vapor **615** and water vapor **710** may then be accumulated in the venting chamber, raising the liquid level in the condensing chamber **430**. A liquid level sensor **577** operatively disposed in the condensing chamber **430** may be configured to measure the liquid level. If the measured liquid level exceeds a preferred liquid level, a control device may be adapted to trigger the liquid pump **576**, so that all or some portion of the liquid in the condensing chamber **430** will be pumped into the water separator **565**. One advantage of this alternative embodiment is that with a pump **576**, the water separator **565** no longer needs to be placed at a location with sufficient gravitational potential difference with the condensing chamber **430**.

[0076] In yet another embodiment, the water separator **565** may include a tilted perforated plate **566** in a container **564**, as shown in FIG. **11**. The container **564** may be structured and arranged to include an inlet **571**, a dielectric fluid chamber **567**, a dielectric fluid drain **569**, a water chamber **568**, and a water drain **570**. In some implementations, the water separator **565** may be configured to separate dielectric fluid **620** from water **715** based on properties of the liquids, such as differing surface tensions. For example, HFE-7100 has a typical surface tension of 13.6 dynes/cm, while water has a typical surface tension of 72 dynes/cm. As a result, the dielectric fluid and water mixture may flow across the tilted perforated plate **566**, but due to surface tension differences, the dielectric fluid **620** with a lower surface tension will flow through the perforations or holes **572** into the fluid chamber **567** beneath the tilted perforated plate **566** while the water with higher surface tension will flow to the end of the tilted perforated plate **566** and into the water chamber **568**. While HFE-7100 is used as an example, such properties can be exploited with any kind of dielectric fluid that has a different kinematic viscosity than water.

[0077] The size of perforations or holes **572** in the tilted perforated plate **566** may vary based on the surface tension of the dielectric fluid. For example, a mesh size of about 60-200 may be effective to separate HFE-7100 from water. A mesh size of 80 implies that there are 80 holes across a square-inch area. A mesh size of 80 may include holes having diameters of about 0.18 mm. In another embodiment, the tilted perforated plate **566** may be replaced with a sieve. The sieve may be made of metal wire. The sieve may have a mesh size of about 60-200.

[0078] The apparatus **100** may also include a filtration system **575**. The filtration system **575** may include a drying filter **590** that is, in some variations, fluidically connected to the liquid return passage **470** and disposed between the outlet **440** of the condensing chamber **430** and the inlet of the immersion tank **201**. Preferably, the drying filter **590** may include a desiccant material.

[0079] The apparatus **100** may be constructed from metal, such as carbon steel. The immersion tank **201** may be constructed of metal with welded seams. Metal materials may be preferable over plastic materials, since metal materials may effectively prevent moisture transfer from the ambient environment to the dielectric fluid **620** in the tank **201**. Minimizing moisture transfer to the dielectric fluid **620** is desirable to reduce dewatering demand placed on the water separator **565** and may also reduce dewatering demand on the desiccant material in the drying filter **590**.

[0080] In some applications, the filtration system **575** may include an impurity filter **580**. The impurity filter **580** may be fluidically connected to the liquid return passage **470** and disposed between the outlet of the drying filter **590** and the inlet of the immersion tank **201**. The impurity filter **580** may include activated carbon, charcoal, and the like. The impurity filter **580** may capture any impurities or debris.

[0081] In some variations, the apparatus **100** may include a liquid return system. The liquid return system may be configured to return dielectric liquid **620**, that has been condensed from dielectric vapor, to the immersion tank **201**. The liquid return system may include a liquid pump **585**. The liquid pump **585** may be fluidically connected to the liquid return passage **470** and disposed between the outlet **574** of the water separator **565** and the inlet to the immersion tank **201**. For example, the liquid pump **585** may be located upstream of the filters **580**, **590** or, alternatively may be located downstream of the filters **580**, **590**.

[0082] Due to its efficient design, the two-phase immersion cooling apparatus **100** may require significantly less dielectric fluid than competing designs that rely on relatively large internal or external reservoirs of subcooled fluid to function properly. Reducing fluid volume is desirable to reduce fluid cost, system weight, and system size. Minimizing size and weight may be particularly important in mobile applications and stationary applications where an engineered floor is not available to support the apparatus **100**.

[0083] In some embodiments, the vapor management system **400** may only be needed periodically. For instance, when the electronic device **800** idles, operates below its maximum power rating, operates at relatively constant power with little variation, or the like, the vapor management system **400** may not be needed until the electronic device power increases. In other embodiments, the vapor management system **400** may be needed frequently but may have sufficient cooling capacity to serve multiple immersion tanks **201** simultaneously. In either scenario, a central vapor management system **400** may serve two or more immersion tanks **201**, **201'**. FIG. **17** shows an embodiment of a two-phase immersion cooling apparatus **1700** having a central vapor management system **400** fluidically connected to a first immersion tank assembly **200** and a second immersion tank assembly **200'**. Employing a central vapor management system **400** may be less expensive than equipping each immersion tank **201**, **201'** with a separate vapor management system **400**. Employing the central vapor management system **400** may conserve floor space in the data center **2000** (FIG. **4**). Employing the central vapor management system **400** may reduce or simplify maintenance. Each immersion tank **201**, **201'** may be fluidically connected to the vapor management system **400** by a corresponding vapor supply passage **405**, **405'** and may be fluidically connected to the water separation and filtration assembly **500** by a liquid return passage **470**.

[0084] The central vapor management system **400** may monitor tank pressure in each immersion tank **201**, **201'** (e.g., using a (e.g., pressure) sensor(s) **412**) and receive dielectric vapor **615** from one tank **201**, both tanks **201**, **201'**, or neither tank as necessary to maintain tank pressures within acceptable ranges. For example, the central vapor management system **400** may receive vapor from the first immersion tank **201** when the first immersion tank pressure is at or above a first threshold pressure. The central vapor management system **400** may receive vapor from the second immersion tank **201'** when the second immersion tank pressure is at or above a second threshold pressure.

[0085] FIG. **18** shows a two-phase immersion cooling apparatus **1800** that varies from the apparatus **1700** of FIG. **17** by including a vapor pump **420**, **420'** that is fluidically connected between a corresponding immersion tank **201**, **201'** and the central vapor management system **400**. Each vapor pump **420**, **420'** may be configured to purge a mixture of air and dielectric vapor from each immersion tank **201**, **201'** and force it into the condensing chamber **430**. Each vapor pump **420**, **420'** may be located upstream of the condensing chamber **430**. Each vapor pump **420**, **420'** may have an inlet and an outlet. The inlet of each vapor pump **420**, **420'** may be fluidically connected to the outlet of the corresponding variable volume chamber **415**, **415'**. The outlet of each vapor pump **420**, **420'** may be fluidically connected to the inlet **435** of the condensing chamber **430**. Advantageously, each vapor pump **420** may be capable of reducing the pressure in a respective immersion tank below atmospheric pressure.

[0086] In some embodiments, the vapor management system **400** may be integrated into the two-phase immersion cooling apparatus **100** and be located in a common enclosure. In other embodiments, the vapor management system **400** may be included in a separate vapor processing apparatus **900**, as shown in FIGS. **19-22**, that fluidically connects to one or more immersion tank assemblies **200**, **200'**. The vapor processing apparatus **900** may retrofit to an existing two-phase immersion cooling apparatus **100** to boost vapor management capacity. For instance, if electronic devices **800** in the data center **2000** are upgraded and will consume more power than the electronic devices **800** being replaced, the cooling capacity of the two-phase immersion cooling apparatus **100** may need to be upgraded to manage the additional heat load. Rather than replacing the apparatus **100**, alternatively, the vapor management system **400** may be added to the apparatus **100** to manage higher vapor production rates, allowing the existing immersion tank assemblies **200**, **200'** to be reused.

[0087] In some implementations, the vapor processing apparatus **900** may include an enclosure **905**, as shown in FIG. **19**. The vapor processing apparatus **900** may include a vapor management system **400**. The vapor management system **400** may include a vapor supply passage **405** having a vapor supply inlet **401**. The vapor management system **400** may include a condensing chamber **430** having an inlet, an outlet, and an auxiliary condenser **451** in thermal communication with an interior volume of the condensing chamber **430**. The vapor supply passage **405** may fluidically connect the vapor supply inlet **401** to the inlet **435** of the condensing chamber **430**. The vapor management system **400** may include a flow control valve **410** in the vapor supply passage **405** between the vapor supply inlet **401** and the inlet **435** of the condensing chamber **430**. The vapor management system **400** may include a liquid return passage **470** fluidically connecting the outlet **440** of the condensing chamber **430** to the liquid return outlet **471**. The vapor management system **400** may include a variable volume chamber **415** fluidically connected to the vapor supply passage **405** between the vapor supply inlet **401** and the inlet **435** of the condensing chamber **430**. The vapor management system **400** may also include a (e.g., pressure) sensor **412** located in the vapor supply passage **405** and configured to detect pressure within an immersion tank **201**, when fluidically connected. The vapor management system **400** may further include a pressure relief valve **460** fluidically connected to the condensing chamber **430**.

[0088] The vapor processing apparatus **900** may include a water separation and filtration assembly **500**. The vapor processing apparatus **900** may include, as shown in FIG. **10**, a water separator **565** and/or a filtration system **575**. The filtration system **575** may include one or more of: a liquid pump **585**, a drying filter **590**, an impurity filter **580**, and so forth.

[0089] FIG. **19** shows a vapor processing apparatus **900** having a single vapor supply inlet **401**. FIG. **20** shows a vapor processing apparatus **900** having two vapor supply inlets **401**, **401'**. In other embodiments, the vapor processing apparatus **900** may have more than two vapor supply inlets **401**, **401'** to allow the apparatus **900** to receive vapor from more than two immersion tanks **201**, **201'**. In some embodiments, the vapor processing apparatus **900** may receive vapor from a grouping of immersion tanks. For instance, the vapor processing apparatus **900** may receive vapor from the groupings of immersion tanks **201**, **201'** disposed in a plurality of tank assemblies **200**, **200'**, as shown in FIG. **2** or **4**. Having a central vapor processing apparatus **900** may be more efficient and more cost-effective than having a vapor processing apparatus **900** for each immersion tank **200**.

[0090] In some variations, the vapor processing apparatus **900** may include a vapor pump **420** to allow the apparatus **900** to actively purge dielectric vapor **615** and air **705** (FIG. **5**) from a headspace **206** (FIG. **5**) of an immersion tank **200**. FIG. **21** shows a vapor processing apparatus **900** with a single vapor supply inlet **401** and a single vapor pump **420** fluidically connected to the vapor management system **400**. FIG. **22** shows a vapor processing apparatus **900** with two vapor supply inlets **401**, **401'** and corresponding vapor pumps **420**, **420'** fluidically connected to corresponding vapor supply passages **405**, **405'** of the vapor management system **400**.

Methods of Operation

[0091] Prior to use, the immersion tank **201** may be partially filled with liquid dielectric fluid **620**, as shown in FIG. **5**. The remainder of the immersion tank **201** may be filled with air **705** at about atmospheric pressure (e.g., 1 atm). An interface between the liquid dielectric fluid **620** and air **705** may define a liquid line **605**. In some implementations, the fluid **620** may be non-toxic. Furthermore, in some variations, the fluid **620** may be non-conductive and pose no risk to electronic devices.

[0092] Electronic devices **800** that require cooling, such as servers, switches, routers, firewalls, and the like, may be immersed in the fluid **620** within the immersion tank **201**, as shown in FIG. **5**. For example, the electronic devices **800** may be fully immersed and positioned below the liquid line **605**. Power and communication cables (not shown) may extend from the electronic devices **800** to connection locations outside the immersion tank **201**. The cables may pass through an opening in the lid **225** or tank wall. In another embodiment, the immersion tank **201** may include integrated connectors within the tank **201** to simplify cable management. In some applications, the electronic devices **800** may be arranged in a storage rack within the immersion tank **201**.

[0093] Dielectric fluid **620** in the immersion tank **201** may initially be at about room temperature. After the electronic device(s) **800** is immersed in the fluid and powered on, the device **800** may begin generating waste heat as a byproduct of power consumption. The heat may be absorbed by the dielectric fluid **620**. If heat flux from the device **800** is sufficiently high, localized boiling of the dielectric fluid **620** may occur. Boiling may produce vapor bubbles that ascend to the liquid line **605** through buoyancy and enter the headspace **206** of the immersion tank **201**. The dielectric vapor **615** may be denser than air **705** and, consequently, a saturated vapor zone **625** may form atop the liquid line **605**, as shown in FIG. **5**. The layer of vapor may be referred to as a vapor blanket **625**. When a rate of vapor production exceeds a rate of condensation, the depth of the vapor blanket **625** will grow. Eventually, the vapor level will approach the primary condenser **235**. Heat transfer from the vapor **615** to the primary condenser **235** may promote condensing of the vapor **615** to liquid **620**. The condensed liquid **620** will then passively return to the liquid bath.

[0094] When the electronic devices **800** are operating at steady state, the rate of vapor production and rate of condensing by the primary condenser **235** may reach equilibrium, resulting in a relatively constant vapor pressure within the immersion tank **201**. Vapor pressure for HFE-7100 can be calculated using an Antoine equation shown below, where P is vapor pressure and T is temperature in degrees Celsius:

$$\ln P_{vapor} = 22.415 - 3641.9\left(1/(T+273)\right)$$

[0095] When power consumption by the electronic device **800** increases, waste heat and vapor production will also increase. If elevated power consumption persists, at some point, the rate of vapor production may overwhelm the condensing capability of the primary condenser **235**. Vapor pressure in the immersion tank **201** will then begin to rise. When pressure in the immersion tank **201** reaches a predetermined threshold, the valve **410** may open, as shown in FIG. **6**, and vapor **615** will escape from the immersion tank **201** into the vapor management system **400**. As the vapor management system **400** receives dielectric fluid vapor **615**, air **705**, and water vapor **710** from the headspace **206** of the immersion tank **201**, the variable volume chamber **415** will expand, as shown in FIG. **7**.

[0096] FIG. **8** shows a plot of immersion tank **201** pressure and device **800** power consumption versus time. In this embodiment, the electronic device **800** operates at idle or low power for a period of time. During that time, vapor production may be managed entirely by the primary condenser **235**, and the vapor management system **400** may remain in standby mode. Eventually, device power consumption may increase. Increased power consumption may produce more waste heat which, in turn, may produce more dielectric vapor **615**. As vapor production increases, a condensing capacity of the primary condenser **235** may eventually be exceeded. As vapor **615**, air **705**, and water vapor **710** accumulate in the headspace **206**, the immersion tank **201** pressure begins to rise until it reaches a predetermined threshold pressure setting. Upon reaching the predetermined threshold pressure setting, as shown in FIG. **6**, the flow control valve **410** may be actuated (e.g., by signals from the (e.g., pressure) sensor **412**), thereby allowing vapor **615**, air **705**, and water vapor **710** to escape from the immersion tank **201** and enter the vapor management system **400**. Actuation of the flow control valve **410** may cause the vapor management system **400** to switch from standby mode to active mode and the chiller **446** to turn on or reduce its setpoint temperature to a level suitable to condense admitted vapor **615** and water vapor **710**. As vapor **615**, air **705**, and water vapor **710** exit the immersion tank **201**, the tank pressure may decrease. Eventually, the tank pressure may fall below the threshold pressure setting of the flow control valve **410**, causing, as shown in FIG. **5**, the flow control valve **410** to close and trap vapor **615**, air **705**, and water vapor **710** received from the immersion tank **201** in the vapor management system **400**. The vapor **615** and water vapor **710** may then be condensed within the condensing chamber **430**, optionally dewatered in the water separator **565**, and further dried and filtered through the drying filter **590** and impurity filter **580**, respectively, before being returned as dielectric liquid **620** to the immersion tank **201**. The vapor management system **400** may cycle on and off as needed to receive and condense vapor **615** and water vapor **710** from the immersion tank **201** and thereby maintain pressure in the immersion tank **201** at or below a desired operating pressure.

[0097] In one embodiment, it may be desirable to anticipate a rise in vapor production that will necessitate activation of the vapor management system **400**. A rise in vapor production may be anticipated by monitoring device power consumption. When device power consumption rises above a predetermined level or, alternately, above a predetermined level for a predetermined duration, the vapor management system **400** may switch from standby mode to active mode. Switching from standby mode to active mode may involve activating the chiller **446** or reducing the setpoint temperature of the chiller **446** to a level suitable to condense admitted vapor **615** and water vapor **710**. This advance activation may allow sufficient time for the chiller **446** (FIG. **6**) to reach a desired working temperature before the valve **410** is opened and vapor **615**, air **705**, and water vapor **710** are admitted from the immersion tank **201**. This approach allows for energy conservation when the vapor management system **400** is in standby mode, since the chiller temperature and chiller power consumption can be reduced.

[0098] In some instances, an unexpected heat load may be present within the immersion tank **201** and must be safely mitigated. For example, an unexpected heat load may occur if an electronic device **800** malfunctions and exceeds a maximum power consumption rating. If this occurs, the amount of heat and vapor **615** and water vapor **710** generated may exceed the condensing capacity of the vapor management system **400**. In practice, the flow control valve **410** will open (FIG. **6**) when the tank pressure reaches the predetermined threshold setting of the flow control valve **410**. As vapor **615**, air **705**, and water vapor **710** fill the vapor management system **400**, the bellows **415** may be configured to expand to store additional vapor **615**, air **705**, and water vapor **710**, as shown in FIG. **7**. After the bellows **415** has fully expanded, if vapor production continues to increase, pressure in the vapor management system **400** (e.g., as measured by the (pressure) sensor **463**) will continue to rise. To avoid mechanical failure due to overpressurization, the pressure relief valve **460** may be configured to actuate, as shown in FIG. **9**, when the pressure in the vapor management system **400** exceeds a maximum allowable pressure. Vapor **615**, air **705**, and water vapor **710** will then be discharged to the ambient environment, thereby reducing pressure in the vapor management system **400** and immersion tank **201** and mitigating risk of mechanical failure. Although venting vapor **615**, air **705**, and water vapor **710** to the ambient environment results in fluid loss, which is undesirable, fluid loss is preferable over safety risks associated with over-pressurization.

[0099] As used herein, the term "fluid" may refer to a substance in gas form, liquid form, or a two-phase mixture of gas and liquid. The fluid may be capable of undergoing a phase change from liquid to vapor or vice versa. The liquid may form a free surface that is not created by a container in which it resides, while the gas may not.

[0100] As used herein, the term "vapor" may refer to a substance in a gas phase at a temperature lower than the substance's critical temperature. Therefore, the vapor may be condensed to a liquid by increasing pressure without reducing temperature.

[0101] As used herein, the term "two-phase mixture" may refer to a vapor phase of a substance coexisting with a liquid phase of the substance. When this occurs, a gas partial pressure may be equal to a vapor pressure of the liquid.

[0102] The dielectric fluid **620** may be selected for use in the immersion cooling apparatus based on a variety of factors including operating pressure, temperature, boiling point, cost, or safety regulations that govern installation (e.g., such as regulations set forth in ASHRAE Standard 15 relating to permissible quantities of fluid per volume of occupied building space).

[0103] In some instances, fluid selection may be influenced by desired dielectric properties, desired boiling point, or compatibility with materials used to manufacture the immersion cooling system **100** and electronic devices **800** to be cooled. For instance, the fluid may be selected to ensure little or no permeability through system components and no harmful effects to device **800** components.

[0104] A dielectric fluid **620**, such as a hydrofluorocarbon (HFC) or a hydrofluoroether (HFE), can be used as the fluid in the immersion cooling apparatus **100**. Unlike water, dielectric fluids can be used in direct contact with electronic devices **800**, such as microprocessors **801**, memory modules, power inverters, and the like, without risk of shorting electrical connections.

[0105] Non-limiting examples of dielectric fluids include 1,1,1,3,3-pentafluoropropane (known as R-245fa), hydrofluoroether (HFE), 1-methoxyheptafluoropropane (known as HFE-7000), methoxy-nonafluorobutane (known as HFE-7100). Hydrofluroethers, including HFE-7000, HFE-7100, HFE-7200, HFE-7300, HFE-7500, and HFE-7600, are commercially available as NOVEC Engineered Fluids from 3M Company headquartered in Mapleton, Minnesota. FC-40, FC-43, FC-72, FC-84, FC-770, FC-3283, and FC-3284 are commercially available as FLUOROINERT Electronic Liquids also from 3M Company.

[0106] NOVEC 7100 has a boiling point of 61 degrees C., a molecular weight of 250 g/mol, a critical temperature of 195 degrees C., a critical pressure of 2.23 MPa, a vapor pressure of 27 kPa, a heat of vaporization of 112 KJ/kg, a liquid density of 1510 kg/m3, a specific heat of 1183 J/kg-K, a thermal conductivity of 0.069 W/m-K, and a dielectric strength of about 40 kV for a 0.1 inch gap. NOVEC 7100 works well for certain electronic devices **800**, such as power electronic devices **800** that produce high heat loads and can operate safely at temperatures above about 80 degrees C.

[0107] NOVEC 7100 can be used to cool a microprocessor **801** that has a preferred operating core temperature of about 60-70 degrees C. When the microprocessor **801** is idling and has a surface temperature below 61 degrees C., subcooled NOVEC 7100 near the microprocessor **801** may experience no local boiling. When the microprocessor **801** is fully utilized and has a surface temperature at or above 61 degrees C., NOVEC 7100 may experience vigorous local boiling and vapor production near the microprocessor.

[0108] NOVEC 649 Engineered Fluid is available from 3M Company. It is a fluoroketone fluid (C6-fluoroketone) with a low Global Warming Potential (GWP). It has a boiling point of 49 degrees C., a thermal conductivity of 0.059, a molecular weight of 316 g/mol, a critical temperature of 169 degrees C., a critical pressure of 1.88 MPa, a vapor pressure of 40 kPa, a heat of vaporization of 88 kJ/kg, and a liquid density of 1600 kg/m3.

[0109] NOVEC 649 may be used to cool a microprocessor **801** that has a preferred operating core temperature of about 60-70 degrees C. When the microprocessor **801** is idling and has a surface temperature below 49 degrees C., subcooled NOVEC 649 near the microprocessor **801** may experience no local boiling. When the microprocessor **801** is fully utilized and has a surface temperature at or above 49 degrees C., NOVEC 649 may experience vigorous local boiling and vapor production near the microprocessor **801**.

[0110] The elements and method steps described herein can be used in any combination whether explicitly described or not. All combinations of method steps as described herein can be performed in any order, unless otherwise specified or clearly implied to the contrary by the context in which the referenced combination is made.

[0111] As used herein, the singular forms "a," "an," and "the" include plural referents unless the content clearly dictates otherwise.

[0112] As used herein, the unit "psig" represents gauge pressure in pounds per square inch. A positive value indicates a pressure above atmospheric pressure. A negative value indicates a pressure below atmospheric pressure.

[0113] As used herein, the term "fluidically connected" can describe a first component directly connected to a second component or a first component indirectly connected to a second component by way of one or more intervening components, where fluid, in gas form, liquid form, or a two-phase mixture, may pass from the first component to the second component without escaping to the atmosphere.

[0114] Numerical ranges as used herein are intended to include every number and subset of numbers contained within that range, whether specifically disclosed or not. Further, these numerical ranges should be construed as providing support for a claim directed to any number or subset of numbers in that range. For example, a disclosure of 1-10 should be construed as supporting a range of from 2 to 8, from 3 to 7, from 5 to 6, from 1 to 9, from 3.6 to 4.6, from 3.5 to 9.9, and so forth.

[0115] The methods and compositions of the present invention can comprise, consist of, or consist essentially of the structural elements and process step limitations described herein, as well as any additional or optional steps, components, or limitations described herein or otherwise useful in the art.

[0116] It is understood that the invention is not confined to the particular construction and arrangement of parts herein illustrated and described, but embraces such modified forms thereof as come within the scope of the claims.

[0117] The foregoing description has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the claims to the embodiments disclosed. Other modifications and variations may be possible in view of the above teachings. The embodiments were chosen and described to explain the principles of the invention and its practical application to enable others skilled in the art to best utilize the invention in various embodiments and various modifications as are suited to the particular use contemplated. It is intended that the claims be construed to include other alternative embodiments of the invention except insofar as limited by the prior art.

What is claimed is:

1. A modular data center comprising:

a shipping container;

an immersion tank positioned within the shipping container, the immersion tank comprising an upper portion, a lower portion, and a primary condenser in thermal communication with an interior volume of the immersion tank; and

a vapor management system fluidly connected to the immersion tank, the vapor management system comprising:

a condensing chamber having an inlet, an outlet, and an auxiliary condenser in thermal communication with an interior volume of the condensing chamber;

a vapor supply passage fluidly connecting the upper portion of the immersion tank to the inlet of the condensing chamber;

a valve in the vapor supply passage between the upper portion of the immersion tank and the inlet of the condensing chamber; and

a liquid return passage fluidly connecting the outlet of the condensing chamber to the immersion tank.

2. The modular data center of claim 1, further comprising an external heat rejection system positioned outside the shipping container and fluidly connected to the primary condenser.

3. The modular data center of claim 2, wherein the external heat rejection system is an evaporative cooling tower.

4. The modular data center of claim 2, wherein the external heat rejection system is a dry cooler.

5. The modular data center of claim 1, further comprising a variable volume chamber fluidly connected to the vapor supply passage between the upper portion of the immersion tank and the inlet of the condensing chamber, wherein the condensing chamber is a fixed volume condensing chamber.

6. The modular data center of claim 1, further comprising a pressure relief valve fluidly connected to the condensing chamber.

7. The modular data center of claim 1, further comprising a pressure sensor located in the immersion tank or in the vapor supply passage and configured to detect pressure within the immersion tank.

8. The modular data center of claim 1, further comprising:

an exhaust passage fluidly connected to the condensing chamber;

a pressure relief valve in the exhaust passage; and

a pressure sensor located in the vapor management system and configured to detect pressure within the vapor management system.

9. The modular data center of claim 1, further comprising a water separator fluidly connected to the liquid return passage between the outlet of the condensing chamber and the immersion tank.

10. The modular data center of claim 1, further comprising:

a liquid pump fluidly connected to the liquid return passage between the outlet of the condensing chamber and the immersion tank;

a drying filter fluidly connected to the liquid return passage between the liquid pump and the immersion tank; and

an impurity filter fluidly connected to the liquid return passage between the drying filter and the immersion tank.

11. A modular data center comprising:

a shipping container; and

a two-phase immersion cooling apparatus positioned within the shipping container, the two-phase immersion cooling apparatus comprising:

an immersion tank comprising an upper portion, a lower portion, and a primary condenser in thermal communication with an interior volume of the immersion tank; and

a vapor management system fluidly connected to the immersion tank, the vapor management system comprising:

a condensing chamber having an inlet, an outlet, and an auxiliary condenser in thermal communication with an interior volume of the condensing chamber;

a vapor supply passage fluidly connecting the upper portion of the immersion tank to the inlet of the condensing chamber;

a solenoid valve in the vapor supply passage between the upper portion of the immersion tank and the inlet of the condensing chamber;

a liquid return passage fluidly connecting the outlet of the condensing chamber to the immersion tank;

a pressure sensor configured to detect pressure in the immersion tank; and

an electronic control unit configured to receive a pressure signal from the pressure sensor and send a command signal to the solenoid valve.

12. The modular data center of claim 11, further comprising an external heat rejection system positioned outside the shipping container and fluidly connected to the primary condenser.

13. The modular data center of claim 12, wherein the external heat rejection system is an evaporative cooling tower or a dry cooler.

14. The modular data center of claim 11, wherein the condensing chamber has a volume at least 10% as large as a headspace volume of the immersion tank.

15. The modular data center of claim 11, wherein the vapor management system further comprises a chiller fluidly connected to the auxiliary condenser.

16. A method comprising:

providing a modular data center comprising:

a shipping container;

a two-phase immersion cooling apparatus positioned within the shipping container, the two-phase immersion cooling apparatus comprising:

an immersion tank comprising an upper portion, a lower portion, and a primary condenser in thermal communication with an interior volume of the immersion tank; and

a vapor management system comprising:

a condensing chamber having an inlet, an outlet, and an auxiliary condenser in thermal communication with an interior volume of the condensing chamber;

a vapor supply passage fluidly connecting the upper portion of the immersion tank to the inlet of the condensing chamber;

a valve in the vapor supply passage between the upper portion of the immersion tank and the inlet of the condensing chamber; and
a liquid return passage fluidly connecting the outlet of the condensing chamber to the immersion tank; and
an external heat rejection system positioned outside the shipping container and fluidly connected to the primary condenser;
detecting a pressure within the immersion tank;
opening the valve when the pressure is greater than a predetermined threshold setting and admitting dielectric vapor from the immersion tank into the condensing chamber;
closing the valve when the pressure in the immersion tank is less than the predetermined threshold setting;
condensing the dielectric vapor to a liquid in the condensing chamber; and
returning the liquid to the immersion tank through the liquid return passage.

17. The method of claim **16**, further comprising circulating a coolant through the primary condenser and external heat rejection system, the coolant having a temperature greater than or equal to an ambient air temperature.

18. The method of claim **16**, further comprising:
circulating a first coolant from the primary condenser to the external heat rejection system; and
circulating a second coolant through the auxiliary condenser,
wherein a temperature of the first coolant is greater than a temperature of the second coolant.

19. The method of claim **16**, further comprising:
providing a vapor pump fluidly connected to the vapor supply passage between the upper portion of the immersion tank and the inlet of the condensing chamber; and
operating the vapor pump while the valve is open to purge gas from a headspace of the immersion tank and reduce the pressure within the immersion tank below atmospheric pressure.

20. The method of claim **16**, wherein the predetermined threshold setting is between −0.9 psig and 0.9 psig.

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